國立政治大學應用數學系

碩士學位論文

離散時間鞅及其在卡爾曼濾波之應用研究

A Study of Discrete Time Martingales with Applications in the Kalman Filter

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中文摘要

在此篇論文中介紹了鞅理論與卡爾曼濾波器的關聯，可以做為探索應用機率與隨機控制的入門。從一些關於鞅的定義以及基本性質開始，接著闡述這些機率的觀念如何應用在卡爾曼濾波上。當系統有噪音干擾時，卡爾曼濾波器是一個最基本的方法可以提供對於系統狀態的估計。接著介紹一些隨機控制的內容，特別是卡爾曼濾波提供的估計可以視作一個具體的應用在線性二次高斯問題（LQG) 上。此篇論文為初學者提供一個清晰、易懂的基礎，連結機率論的核心概念與濾波和控制的應用。

關鍵字：鞅理論、卡爾曼濾波器、遞迴估計、隨機過程、控制理論、線性二次高斯（LQG）問題、應用機率、隨機控制。

Abstract

This thesis serves as an introductory guide for beginners in applied probability and stochastic control, focusing on the connection between martingale theory and the Kalman filter. By starting with the basics of martingales, the thesis explains how these probabilistic concepts can be applied to understand the Kalman filter, a fundamental tool for estimating the state of a system in the presence of noise. The thesis then introduces control theory, specifically the Linear Quadratic Gaussian (LQG) problem, to demonstrate the practical use of the Kalman filter in optimizing system performance. This work aims to provide a clear and accessible foundation for those new to these topics, linking key ideas in probability with their applications in filtering and control.

Keywords: Martingale theory, Kalman filter, Recursive estimation, Stochastic processes, Control theory, Linear Quadratic Gaussian (LQG) problem, Applied probability, Stochastic control.

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Chapter 1

Introduction

Background and Context:

The study of probability, particularly martingale theory, plays a critical role in understanding complex stochastic processes. Martingales provide powerful tools for analyzing convergence properties and other aspects of random processes, which are foundational in various applications, including filtering and control theory. One of the key applications of martingale theory is the Kalman filter, a widely used method for estimating the state of a dynamic system in the presence of noise.

The Kalman filter is not only fundamental in the field of filtering but also has significant applications in control theory. Specifically, in the Linear Quadratic Gaussian (LQG) problem, the Kalman filter is used to achieve optimal control in systems where both state estimation and control are required. Understanding the connections between martingale theory and the Kalman filter provides valuable insights into how probabilistic methods can be applied to solve real-world problems in engineering, finance, and other fields.

Purpose and Objectives:

This thesis aims to serve as an introduction for readers who are already familiar with probability and real analysis, guiding them through the relationship between martingale theory

and the Kalman filter. The primary objectives are:

- To provide a clear and accessible explanation of martingale theory, emphasizing its relevance to stochastic processes and filtering problems.
- To demonstrate how the Kalman filter applies these probabilistic concepts to achieve accurate state estimation in noisy environments.
- To introduce control theory, particularly the LQG problem, as a practical context where the Kalman filter is used to optimize system performance.

Scope of the Thesis:

The thesis assumes that readers have a background in probability and real analysis. It is structured to guide readers from the foundational concepts of martingale theory to the practical applications of the Kalman filter in control theory. While the focus is on summarizing and synthesizing existing knowledge, the work also highlights the practical importance of these concepts in applied probability and stochastic control.

Overview of the Thesis Structure:

- Chapter 2: introduces martingale theory, explaining key concepts and their relevance to stochastic processes.
- Chapter 3: presents the Kalman filter, showing how it builds on martingale theory to solve filtering problems.
- Chapter 4: discusses the application of the Kalman filter in control theory, focusing on the Linear Quadratic Gaussian (LQG) problem.
- Chapter 5: concludes with a summary of the key findings and suggests directions for further research.

Chapter 2

Martingales

2.1 Martingales

This chapter is written with reference to *part B: MARTINGALE THEORY* in David Williams's *Probability with Martingales* (1991) [18].

A discrete-time martingale is a sequence of random variables that captures the idea of a "fair game" in probabilistic terms. It is a central concept in probability theory that plays a crucial role in understanding the behavior of stochastic processes over time, with applications in financial markets, gambling strategies, and various areas such as filtering and control theory. To formalize these concepts, we begin by introducing some essential definitions and properties that will underpin the subsequent discussion of martingales, submartingales, and supermartingales.

We consider a probability space $(\Omega, \mathcal{F}, P)$.

Definition 2.1.1. *A **filtration** $(\mathcal{F}_n)_{n\geq 0}$ on $(\Omega, \mathcal{F}, P)$ is an increasing family of sub-σ-algebra of $\mathcal{F}$ such that*

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq ... \subseteq \mathcal{F}_\infty \subseteq \mathcal{F}$$

, where $\mathcal{F}_\infty \coloneqq \sigma(\bigcup_{n=0}^{\infty} \mathcal{F}_n)$.

*As usual, if $X = (X_n)_{n\geq 0}$ is a random process, we take $\mathcal{F}_n^X = \sigma(X_0, X_1, ..., X_n)$ for every $n \geq 0$ and called it the **canonical filtration** of X.*

*A **filtered space** $(\Omega, \mathcal{F}, (\mathcal{F}_n), P)$ is a probability space equipped with filtration $(\mathcal{F}_n)_{n\geq 0}$ of its σ-algebra $\mathcal{F}$.*

With these preliminaries in place, we can now define what it means for a random process to be adapted to a given filtration.

Definition 2.1.2. *A random process $X = (X_n)_{n\geq 0}$ is said to be **adapted** to the filtration $(\mathcal{F}_n)_{n\geq 0}$ if for each n, X_n is $\mathcal{F}_n$-measurable.*

Adapted processes are those that respect the natural filtration of the system, meaning that the value of the process at any time n is determined by the information available up to that time. This concept is crucial in the definition of martingales, as it ensures that the process evolves in a manner consistent with the accumulation of information over time.

Definition 2.1.3. *A random process X is a **martingale** relative to $(\Omega, \mathcal{F}, (\mathcal{F}_n), P)$ if*

1. *X is adapted,*
2. *X_n is integrable, for $n \geq 0$ (i.e. $E|X_n| < \infty$, for $n \geq 0$)*
3. *$E(X_n|\mathcal{F}_{n-1}) = X_{n-1}$, for $n \geq 1$*

At each step in a martingale process, the expected future value of the process, given all the information up to the present moment (which includes the current value), is exactly equal to the current value. This property implies that there is no predictable trend in the process—no expected gain or loss—making martingales a valuable tool for modeling scenarios where future outcomes are uncertain and entirely dependent on the present state.

Definition 2.1.4. *A random process X is a **supermartingale (submartingale)** relative to $(\Omega, \mathcal{F}, (\mathcal{F}_n), P)$ if*

1. *X is adapted,*
2. *X_n is integrable, for $n \geq 0$ (i.e. $E|X_n| < \infty$ for $n \geq 0$)*

3-i ***(supermartingale)*** *$E(X_n|\mathcal{F}_{n-1}) \leq X_{n-1}$ for $n \geq 1$*

3-ii ***(submartingale)*** $E(X_n|\mathcal{F}_{n-1}) \geq X_{n-1}$ *for* $n \geq 1$

Supermartingales and submartingales generalize the concept of martingales, allowing for systematic downward or upward drifts, respectively.

Remark 1. *We refer to the condition 3 as the martingale property.*

To solidify the understanding of martingales, let's consider some illustrative examples.

Example 1 (Sums of Independent Centered Random variables)**.** *Let* $X_1, X_2, \ldots$ *be a sequence of integrable indepedent random variables with* $E[X_i] = 0 \quad \forall i \geq 1$.

Define

$$S_n := \sum_{i=1}^{n} X_i \quad \textit{and} \quad S_0 := 0$$

$$\mathcal{F}_n := \sigma(X_i : 1 \leq i \leq n) \quad \forall n \geq 0. \quad \textit{and} \quad \mathcal{F}_0 := \{\phi, \Omega\}.$$

Check for conditions of martingale:

1. $S_n = \sum_{i=1}^{n} X_i \in \mathcal{F}_n$ *, since each* $X_i \in \mathcal{F}_i$ *and* $(\mathcal{F}_i)_{i\geq 0}$ *is increasing.*
2. $E|S_n| = E|\sum_{i=1}^{n} X_i| \leq E\sum_{i=1}^{n} |X_i| = \sum_{i=1}^{n} E|X_i| < \infty$.
3. *For* $n \geq 1$*,*

$$\begin{aligned} E[S_n|\mathcal{F}_{n-1}] &= E[S_{n-1} + X_n|\mathcal{F}_{n-1}] = E[S_{n-1}|\mathcal{F}_{n-1}] + E[X_n|\mathcal{F}_{n-1}] \\ &= S_{n-1} + E[X_n] \\ &= S_{n-1} \end{aligned}$$

Since $E[S_{n-1}|\mathcal{F}_{n-1}] = S_{n-1}$ *(For* S_{n-1} *is measurable to*$\mathcal{F}_{n-1}$*) and* $E[X_n|\mathcal{F}_{n-1}] = E[X_n] = 0$ *(For*X_n *is independent of* $\mathcal{F}_{n-1}$ *).*

Next, we consider another fundamental example of martingales that involves conditional expectations, demonstrating the role of martingales in updating information over time.

Example 2 (Accumulating Information)**.** *Let $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$ be an integrable random variable defined on $(\Omega, \mathcal{F}, P)$ and $(\mathcal{F}_n)_{n\geq 0}$ be a filtration.*

Define

$$M_n := E[\xi|\mathcal{F}_n] \quad \forall n \geq 0.$$

Check for conditions of martingale:

1. *$M_n = E[\xi|\mathcal{F}_n]$ is measurable to $\mathcal{F}_n$ for each $n \geq 1$, by definition of conditional expectation i.e. $(M_n)_{n\geq 0}$ is adapted.*
2. *$E|M_n| = E|E(\xi|\mathcal{F}_n)| \leq EE[|\xi||\mathcal{F}_n] = E|\xi| < \infty$, since ξ is integrable i.e. each M_n is integrable.*
3. *For $n \geq 1$, $E[M_n|\mathcal{F}_{n-1}] = E[E(\xi|\mathcal{F}_n)|\mathcal{F}_{n-1}] = E[\xi|\mathcal{F}_{n-1}] = M_{n-1}$ (the second equality holds for tower property.)*

So $(M_n)_{n\geq 0}$ is a martingale.

Hence, $(M_n)_{n\geq 0}$ is a martingale. This example is particularly important because it can be interpreted as accumulating information about a random variable over time, a concept that will be frequently used in subsequent sections.

In the study of stochastic processes, it is often important to identify specific times at which certain events occur. These moments, however, are not always fixed in advance but can depend on the evolution of the process itself. To formalize this idea, we introduce the concept of a stopping time, which allows us to pinpoint the exact time at which a particular event happens, based on the information available up to that point.

Definition 2.1.5 (Stopping Time)**.** *Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration. A random variable $T : \Omega \mapsto \{0, 1, 2, \ldots, \infty\}$ is a **stopping time** with respect to $(\mathcal{F}_n)_{n\geq 0}$ if*

$$\{T = n\} = \{\omega \in \Omega : T(\omega) = n\} \in \mathcal{F}_n \quad \forall n \leq \infty.$$

Example below illustrates the concept of a stopping time, specifically in the context of a process reaching a particular value, often referred to as the ”first hitting time.”

Example 3 (The First Hitting Time)**.** *Suppose* $(X_n)_{n\geq 1}$ *is an i.i.d. sequence of random variables on* $(\Omega, \mathcal{F}, P)$ *with* $P(X_n = \pm 1) = \frac{1}{2}$.

Define

$$S_n := S_0 + \sum_{i=1}^{n} X_i \quad \textit{and} \quad S_0 := 5.$$

$$\mathcal{F}_n := \sigma(S_i : 1 \leq i \leq n) \quad \textit{and} \quad \mathcal{F}_0 = \{\phi, \Omega\}.$$

Let

$$T = \inf\{n \geq 0 : S_n = 10 \quad \textit{or} \quad S_n = 0\}.$$

Check the definition that T *is a stopping time with respect to* $(\mathcal{F}_n)_{n\geq 0}$ *:*
For each n*,*

$$\{T = n\} = \bigcap_{i=1}^{n-1} \{0 < S_i < 10\} \bigcap \{S_n = 10 \quad \textit{or} \quad S_n = 0\}$$

where for $1 \leq i \leq n-1$, $\{0 < S_i < 10\} \in \mathcal{F}_i \quad (\subseteq \mathcal{F}_n)$ *and*
$\{S_n = 10 \quad \textit{or} \quad S_n = 0\} \in \mathcal{F}_n$.

So $\{T = n\} \in \mathcal{F}_n$*, thus* T *is a stopping time.*

This example demonstrates how stopping times can be used to identify the first moment at which a process reaches a particular value.

2.2 Almost Sure Convergence of Martingales

Understanding the long-term behavior of martingales is both natural and essential. A key question is whether a martingale converges almost surely. To explore this, we introduce the concept of upcrossings, which counts the number of times a martingale crosses a given interval. This is fundamental in establishing whether and how the process converges.

2.2.1 Upcrossings

The upcrossing number measures how many times a martingale crosses a given interval.

To define the sequence of stopping times (s_k, t_k) that specify the upcrossing number of a martingale, proceed as follows:

Let $a < b$.

- s_1 is the first time such that $X_n < a$.

  i.e. a stopping time

  $s_1 = \inf\{n \geq 0 : X_n < a\}$, if there is $n \geq 0$ such that $X_n < a$.

  Otherwise define $s_1 = \infty$.

- t_1 is the first time after s_1 such that $X_n > b$.

  i.e. a stopping time

  $t_1 = \inf\{n \geq s_1 : X_n > b\}$, if $s_1 < \infty$ and there is $n \geq s_1$ such that $X_n > b$.

  Otherwise define $t_1 = \infty$.

- s_2 is the first time after t_1 such that $X_n < a$.

  i.e. a stopping time

  $s_2 = \inf\{n \geq t_1 : X_n < a\}$, if $t_1 < \infty$ and there is $n \geq t_1$ such that $X_n < a$.

  Otherwise define $s_2 = \infty$.

- t_2 is the first time after s_2 such that $X_n > b$

  i.e. a stopping time

  $t_2 = \inf\{n \geq s_2 : X_n > b\}$, if $s_2 < \infty$ and there is $n \geq s_2$ such that $X_n > b$.

  Otherwise define $t_2 = \infty$.

  And so on.

Then for $k \geq 0$, define $t_0 = 0$ and inductively, we define,

$s_{k+1} := \inf\{n \geq t_k : X_n < a\}$,

if $t_k < \infty$ and there is $n \geq t_k$ such that $X_n < a$. Otherwise define $s_{k+1} = \infty$.

$t_{k+1} := \inf\{n \geq s_{k+1} : X_n > b\}$,

if $s_{k+1} < \infty$ and there is $n \geq s_{k+1}$ such that $X_n > b$. Otherwise define $t_{k+1} = \infty$.

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Definition 2.2.1 (Upcrossing Number)**.** *The upcrossing number of* $(X_n)_{n\geq 1}$ *along* $[a,b]$ *up to time* N*, denoted by* $U_N[a,b](\omega)$*, is the largest* k *such that* $t_k \leq N$*. Then*

$$U_N[a,b](\omega) := \max\{k \geq 1 : t_k \leq N\}.$$

The upcrossing number is not just a counting tool; it also provides insights into the behavior of supermartingales. Doob's Upcrossing Lemma gives us a bound on the expected number of upcrossings a supermartingale can make.

Lemma 2.2.1 (Doob's Upcrossing Lemma)**.** *Let* $(X_n)_{n\geq 0}$ *be a supermartingale. Let* $U_N[a,b]$ *be the upcrossing number of* $(X_n)_{n\geq 1}$ *along* $[a,b]$ *up to time* N*. Then*

$$(b-a)EU_N[a,b] \leq E[(X_N - a)^-].$$

Corollary 1. *Let* $(X_n)_{n\geq 0}$ *be a supermartingale bouded in* $\mathcal{L}^1$ *i.e.* $\sup_{n\geq 0} E|X_n| < \infty$*. Write* $U_\infty[a,b] := \lim_{N\to\infty} U_N[a,b]$*, the limit exists in* $[0,\infty]$*.* *(*$\because U_N[a,b]$ *is increasing in* N*.)*
Then

$$U_\infty[a,b] = \infty \quad \textit{with probability 0.}$$

Building on this, we now consider Doob's Convergence Theorem. This theorem formalizes the idea that, under certain conditions, not only does the upcrossing number remain finite, but the martingale itself converges to a finite limit almost surely as time progresses.

Theorem 2.2.1 (Doob's Convergence Theorem)**.** $(X_n)_{n\geq 0}$ *is a* $\mathcal{L}^1$*-bounded martingale. Then*

$$X_\infty := \lim_{n\to\infty} X_n \quad \textit{exists and is finite a.s..}$$

i.e. $\exists F(\subseteq \Omega)$ *with* $P(F) = 0$ *such that* $X_\infty(\omega)$ *exists and is finite* $\forall \omega \in \Omega \setminus F$*.*
Also for $\omega \in N$ *set* $X_\infty(\omega) := \limsup X_n(\omega)$*.*

The proof of Doob's Convergence Theorem is omitted here, but to whom is interested can refer to David Williams's *Probability with Martingales* (1991, pp. 109) [18].

2.3 Uniformly Integrable Martingales

While almost sure convergence is a powerful result, in many applications, it is also important to strengthen this result by proving $\mathcal{L}^1$- convergence of martingales. This is the focus of our discussion in this section, where we aim to extend Doob's Convergence Theorem to achieve convergence in the $\mathcal{L}^1$ sense.

Remark 2 (Clarify $\mathcal{L}^p$ Terms)**.** *For $p \in [1, \infty)$,*

- $(X_n)_{n\geq 0}$ *is in $\mathcal{L}^p$ (or $\mathcal{L}^p$-integrable) , if* $E[|X_n|^p] < \infty$ *for all* $n \geq 0$.
- $(X_n)_{n\geq 0}$ *is bounded in $\mathcal{L}^p$ (or $\mathcal{L}^p$-bounded), if* $\sup_{n\geq 0} E|X_n|^p < \infty$*, or equivalently for some* $A \in [0, \infty)$, $E|X_n|^p < A$, $\forall n \geq 0$.

2.3.1 Uniformly integrable property

To achieve $\mathcal{L}^1$- convergence, a key concept is that of uniform integrability (UI), which we now introduce. The following lemma provides a foundational property of uniformly integrable random variables.

Lemma 2.3.1. *Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$ be an integrable random variable. Given $\epsilon > 0$, there exists $\delta > 0$ s.t.*

$$P(F) < \delta \Rightarrow E\big[|X|I_F\big] < \epsilon.$$

Proof. Suppose contrary, then for $\epsilon_0 > 0$ we can find $(F_n)_{n\geq 0} \in \mathcal{F}$ such that $P(F_n) < \frac{1}{2^n}$ and $E[|X|I_{F_n}] > \epsilon_0$.

$\sum P(F_n) = \frac{1}{2} < \infty$ then by Borel – Cantelli lemma $P(\limsup_n F_n) = 0$. So $E[|X|I_{\limsup_n F_n}] = 0$. By Reverse Fatou lemma, we get

$$E[|X|I_{\limsup_n F_n}] \geq \limsup_n E[|X|I_{F_n}] \geq \epsilon_0.$$

Hence we get the contradition. □

This result is a key step in understanding the behavior of integrable random variables over sets of small probability, which leads directly to the next important lemma.

Lemma 2.3.2. *Let $X \in \mathcal{L}^1$ be an integrable random variable and $\epsilon > 0$. Then there exists $K \in [0, \infty)$ such that*

$$E\big[|X| I_{\{|X|>K\}}\big] < \epsilon.$$

Moreover,

$$\lim_{K\to\infty} E\big[|X| I_{\{|X|>K\}}\big] = 0$$

which means

$$E\big[|X| I_{\{|X|>K\}}\big] \to 0 \text{ as } K \to \infty.$$

Proof. By Markov inequality $P(|X| \geq K) \leq \dfrac{E|X|}{K}$.
Choose K s.t. $P(|X| \geq K) < \delta$ then $E\big[|X| I_{\{|X|\geq K\}}\big] < \epsilon$ from the last lemma. □

These results naturally lead us to the concept of uniform integrability, which is crucial for $\mathcal{L}^1$ convergence in martingales. We now formalize this concept in the following definition.

Definition 2.3.1 (Uniformly Integrable)**.** *Let $\mathcal{C}$ be a collection of integrable random variables. Then the collection $\mathcal{C}$ is said to be **uniformly integrable (UI)**, if for any $\epsilon > 0$, there is $K \in [0, \infty)$ such that*

$$E\big[|X| I_{\{|X|>K\}}\big] < \epsilon, \ \forall X \in \mathcal{C}.$$

That is, the collection $\mathcal{C}$ satisfies the following property:

$$\lim_{K\to\infty} \Big[\sup_{X\in\mathcal{C}} E\big[|X| I_{\{|X|>K\}}\big] \Big] = 0,$$

meaning

$$\sup_{X\in\mathcal{C}} E\big[|X| I_{\{|X|>K\}}\big] \to 0 \text{ as } k \to \infty.$$

This definition encapsulates the idea that the tail behavior of random variables in a uniformly integrable collection is uniformly negligible, which is a powerful concept in proving $\mathcal{L}^1$ convergence.

Property 2.3.1. *If a collection of integrable random variables is uniformly integrable (UI), then it is also $\mathcal{L}^1$-bounded.*

Proof. Let $\mathcal{C}$ be a UI collection of integrable random variables. Suppose K is the response to $\epsilon = 1$.

For all $X \in \mathcal{C}$,

$$\begin{aligned} E|X| &= E[|X|I_{\{|X|>K\}}] + E[|X|I_{\{|X|\leq K\}}] \\ &< 1 + KP(|X| \leq K) \\ &< 1 + K \end{aligned}$$

Then $\sup_{X\in\mathcal{C}} E|X| \leq 1 + K < \infty$ i.e. $\mathcal{C}$ is $\mathcal{L}^1$-bounded. □

This property establishes a crucial link between uniform integrability and $\mathcal{L}^1$-boundedness, showing that if a collection of random variables is uniformly integrable, its expected values are uniformly bounded.

A classic example of a UI collection is provided by the class of conditional expectations, as shown below.

Example 4. *Let $X \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$. The collection*

$$\mathcal{C} = \{E[X|\mathcal{G}] : \mathcal{G} \quad \textit{is a sub-}\sigma\textit{-algebra of } \mathcal{F}\}$$

is UI.

Proof. Recall the condition in lemma 2.3.1. Starting by select a small $\delta > 0$ such that for any $F \in \mathcal{F}$ with $P(F) < \delta$, we have $E[|X|I_F] < \epsilon$ for a given $\epsilon > 0$. We then choose a sufficiently large constant K such that $\frac{E|X|}{K} < \delta$.

Let $Y = E[X \mid \mathcal{G}]$. By conditional version of Jensen's inequality,

$$|Y| = \big|E[X \mid \mathcal{G}]\big| \leq E[|X| \mid \mathcal{G}]$$

, and we get

$$E|Y| \leq EE[|X| \mid \mathcal{G}] = E|X|.$$

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It follows that

$$\begin{aligned} P(|Y| > K) &\leq \frac{E|Y|}{K} \\ &\leq \frac{E|X|}{K} < \delta. \end{aligned}$$

Since $\{|Y| > K\} \in \mathcal{G}$, the definition of conditional expectation implies that

$$\begin{aligned} E[|Y|I_{|Y|\geq K}] &\leq EE[|X|I_{|Y|\geq K} \mid \mathcal{G}] \\ &= E[|X|I_{|Y|\geq K}] \\ &< \epsilon. \end{aligned}$$

Hence, the class of conditional expectations is uniformly integrable. □

This example highlights an important instance of a uniformly integrable collection. The collection $\mathcal{C}$, consisting of conditional expectations of X given sub-σ-algebras of $\mathcal{F}$, demonstrates how uniform integrability naturally arises in the context of martingales and conditional expectations, reinforcing the concepts discussed earlier.

To further explore the concept of uniform integrability, we introduce two sufficient conditions that ensure a collection of random variables is uniformly integrable.

Theorem 2.3.1 ($\mathcal{L}^p$ Boundedness)**.** *If $\mathcal{C}$ is a collection of random variables that are bounded in $\mathcal{L}^p$ for some $p \in [0, \infty)$.*

That is, if

$$\sup_{X\in\mathcal{C}} E|X|^p < \infty,$$

or equivalently, if there exists some $A \in [0, \infty)$ such that $E|X|^p < A$ for all $X \in \mathcal{C}$, then $\mathcal{C}$ is UI.

This theorem shows that if the random variables in a collection are bounded in $\mathcal{L}^p$ for any $p \geq 1$, the collection is uniformly integrable. This condition is particularly useful in practice, as L^p boundedness is often easier to verify than uniform integrability directly.

Theorem 2.3.2. *If $\mathcal{C}$ is a collection of random variables that are dominated by a non-negative integrable random variable Y.*

That is

$$|X(\omega)| \leq Y(\omega),\ \forall X \in \mathcal{C} \text{ and } E[Y] < \infty.$$

Then $\mathcal{C}$ is UI.

This second theorem leverages the Dominated Convergence Theorem (DCT), showing that if all the random variables in a collection are bounded above by a single integrable random variable Y, then the collection is uniformly integrable. This is a powerful condition, as it provides a straightforward method to establish uniform integrability when an appropriate dominating function can be identified.

The proof of these two sufficient conditions can be found in David Williams's Probability with Martingales (1991, pp. 127–128) [18].

2.3.2 Convergence of UI martingale

In this section, we explore the convergence properties of uniformly integrable martingales, building on the characterization of $\mathcal{L}^1$- convergence. The theorem below not only provides a crucial link between uniform integrability and $\mathcal{L}^1$- convergence but also offers an improvement over the Dominated Convergence Theorem (DCT). This is because the condition of uniform integrability (UI) is often easier to verify in probability theory, especially in the context of martingales, making it a more practical tool for proving convergence results.

Theorem 2.3.3 (Characterization of $\mathcal{L}^1$- Convergence)**.** *Consider a sequence (X_n) of random variables in $\mathcal{L}^1$ and a random variable $X \in \mathcal{L}^1$. The sequence (X_n) converges to X in $\mathcal{L}^1$ (that is, $E(|X_n - X|) \to 0$) if and only if both of the following conditions are met:*

1. *X_n converges to X in probability,*
2. *The sequence (X_n) is uniformly integrable (UI).*

This theorem provides a crucial link between uniform integrability and $\mathcal{L}^1$ convergence, setting the stage for our discussion of martingales. We now introduce the concept of a closed martingale, which plays an essential role in understanding when a martingale will converge in $\mathcal{L}^1$. The proof can be found in David Williams's *Probability with Martingales* (1991, pp. 131-132) [18].

Definition 2.3.2 (Closed Martingale)**.** *A martingale $(X_n)_{n\geq 0}$ is called a closed martingale if there exists an integrable random variable $\xi \in \mathcal{L}^1$ s.t. for every $n \geq 0$*

$$X_n = E[\xi|\mathcal{F}_n].$$

And say $(X_n)_{n\geq 0}$ is closed by ξ.

The concept of a closed martingale allows us to connect the martingale sequence directly to an integrable random variable, providing a foundation for proving convergence results. The following theorem demonstrates the equivalence between being closed, uniformly integrable, and convergent in both almost sure and $\mathcal{L}^1$ senses. The definition of closed martingale is from Legall's *Brownian Motion, Martingales, and Stochastic Calculus* (2016, p.59) [13]. The following theorem (Theorem 2.3.4)and its proof can also be found there.

Theorem 2.3.4. *$(X_n)_{n\geq 0}$ is a martingale then the following is equivalent.*

1. *$(X_n)_{n\geq 0}$ is closed.*
2. *$(X_n)_{n\geq 0}$ is uniformly integrable.*
3. *$(X_n)_{n\geq 0}$ converges a.s. and in $\mathcal{L}^1$.*

We now turn to a key result known as Levy's Upward Theorem, which further illustrates the connection between closed martingales and their convergence properties.

Theorem 2.3.5 (Levy's Upward Theorem)**.** *Let $\xi \in \mathcal{L}^1(\Omega, \mathcal{F}, P)$ and suppose that $(X_n)_{n\geq 0}$ is closed by ξ i.e. $X_n = E[\xi|\mathcal{F}_n]$. Then*

$$\lim_{n\to\infty} X_n = E[\xi|\mathcal{F}_\infty] \text{ a.s. and in } \mathcal{L}^1.$$

Levy's Upward Theorem plays a crucial role in the convergence analysis of martingales, particularly in the context of conditional expectations. It guarantees that if a martingale is closed by an integrable random variable, then it not only converges almost surely but also in $\mathcal{L}^1$. This convergence result is fundamental when dealing with sequences of conditional expectations, as it ensures the stability of the process over time. The proof is provided in David Williams's Probability with Martingales (1991, p. 135) [18].

This theorem leads us into the Section 3.3.2, where we consider a specific example of martingales arising from noisy observations. In the context of recursive estimation, Levy's Upward Theorem provides the theoretical foundation for understanding the convergence of the best estimator of an unknown variable X given increasing amounts of noisy observations.

Chapter 3

Application: The Kalman Filter

3.1 State-Space Model

State-space models are used to predict and control the behavior of complex systems. They focus on input-output relationships and capture the dynamics of all relevant variables, offering a description of a system's internal state and its evolution over time. This makes them valuable in many fields, including aerospace, chemical engineering, robotics, economics, electrical engineering, and biomedical engineering.

For simplicity, if the system desired to be known is of discrete time and linear, it can be described as follows:

Definition 3.1.1 (Linear State-Space Model)**.**

$$x_{n+1} = A_n x_n + B_n u_n \tag{3.1}$$

$$y_n = C_n x_n \quad , \forall n \geq 1 \tag{3.2}$$

Where

- x_n *is an* d*-dimensional vector, representing the system state at time* n *that cannot be observed directly.*

- y_n *is an* r*-dimensional vector, representing the system output or observations measured at time* n*.*

- *u_n is a m-dimensional vector, representing the control input at time n. It is applied to system in time $[n, n+1)$.*

- *A_n, B_n and C_n are $d \times d, d \times m$ and $r \times n$ matrices, respectively, changing over time.*

*The equation (3.1) is named **the state equation**, describing how the system state evolves over time.*
*While equation (3.2) is **the observation equation**, relating the system state to the outputs.*

Remark 3. *If the coefficients of the system do not change over time (i.e., $A_n = A, B_n = B$ and $C_n = C$ for all $n \geq 1$), the system is said to be **time-invariant**.*

The following definitions and remarks share the same notations and meanings as defined previously, so we will only point out the different ones.

Definition 3.1.2 (Stochastic State-Space Model)**.** *If noise processes $(\epsilon_n)_{n\geq 1}$ and $(\eta_n)_{n\geq 1}$ are added to the state and observation equations, it becomes a linear stochastic state-space model:*

$$X_{n+1} = A_n X_n + B_n u_n + \epsilon_n \tag{3.3}$$

$$Y_n = C_n X_n + \eta_n \quad , \forall n \geq 1 \tag{3.4}$$

Where

- *$(\epsilon_n)_{n\geq 1}$ is a sequence of d-dimensional random vectors, representing **the system noise**.*

- *$(\eta_n)_{n\geq 1}$ is a sequence of r-dimensional random vectors, representing the errors made from measurement. Also called the **observation noise**.*

- *Y_n is an r-dimensional random vector representing the output of the system. Additionally, it serves as a noisy observation of X_n, incorporating observation noise and system noise.*

Remark 4 (Nonlinear State-Space Model)**.** *If the state and observation equations do not satisfy a linear relationship, the system is described in a general manner:*

$$X_{n+1} = f(X_n, u_n, \epsilon_n)$$

$$Y_n = h(X_n, \eta_n)$$

Where $f(\cdot,\cdot,\cdot)$ describes the system dynamics and $h(\cdot,\cdot)$ describes the observation process.

State-space models address two main problems in dynamic systems: the **Filtering Problem** and **Control Problem**. The filtering problem involves estimating the system state X_n from noisy observations Y_n. The well-known example is the Kalman filter, also known as Linear Quadratic Estimation (LQE). The control problem focuses on determining the optimal input u_n to achieve a desired state or output, often solved using the Linear Quadratic Regulator (LQR).

These problems are interconnected, as effective control requires accurate state estimation. The Kalman filter provides these estimates, which are then used by the LQR to compute optimal control actions. This integration forms the Linear Quadratic Gaussian (LQG) framework, combining estimation and control. The Separation Principle simplifies this procedure by allowing the estimation and control problems to be solved solely.

3.2 Filtering Problem

In the practical applications mentioned above, system can be model by the state-space model. It is essential to accurately know the true state of the system, denoted by X. However, the true state cannot be observed directly. Instead, we successively obtain observations Y from the measurement device, which include both observation noise and system noise. The objective of the filtering problem is to "filter out" these noise and use all available information up to the current time n to estimate the system state X_n. This means the filter propagates the conditional density of X_n given the sequence of observations $Y_0, \ldots, Y_n$.

Instead of doing directly, this process consists of two main steps: **Prediction** and **Correction**. In the filtering problem, we often assume zero control input and a zero mean initial state in state-space model for simplicity.

1. **Prediction:** Using the state equation to predict the next state based on the current state and input. This provides the prediction estimator: the conditional density of $X_n|Y_0, \ldots, Y_{n-1}$.

2. **Correction:** Using the observation equation to correct the state estimate based on new measurements. This provides the correction estimator: the conditional density of $X_n|Y_0, \ldots, Y_n$.

This two-step process is performed recursively, continuously updating the state estimate as new observations become available. The calculations of how these steps are performed will be introduced later. With additional assumptions on the noise, we will delve into the Kalman filter case.

3.3 Recursive Estimation: Bayesian Case

In Recursive Least Squares (RLS), the goal is to recursively update estimates of states (parameters) when new observations are sequentially received while minimizing the sum of squared errors. The concepts of RLS are closely related to those used in the Kalman filter, where recursive estimation is also a key component. It provides foundational techniques, such as Bayesian structure, and makes understanding the Kalman filter easier.

The following content, Section 3.3, is arranged according to and follows sections 15.6, 15.7 of David Williams's *Probability with Martingales* [18].

3.3.1 Some Preparations

3.3.1.1 Recursive Nature of Conditional Probabilities

For $F \in \mathcal{F}$, we have $P(F) = E[I_F]$. In the familiar way, define $P(F|\mathcal{G})$ to be a version of $E[I_F|\mathcal{G}]$, where $\mathcal{G}$ is a sub-σ-algebra of $\mathcal{F}$.

$$\text{i.e.} \quad P(F|\mathcal{G})(\omega) = E[I_F|\mathcal{G}](\omega) \quad \text{for almost every } \omega.$$

Definition 3.3.1 (Regular conditional probabilities)**.** *On probability space* $(\Omega, \mathcal{F}, P)$*. Let* $\mathcal{G}$ *be a sub-*σ*-algebra of* $\mathcal{F}$*.*
If there exists a map

$$\mathbb{P}(\cdot, \cdot) : \Omega \times \mathcal{F} \to [0, 1]$$

such that

1. *for $F \in \mathcal{F}$, the section $\omega \mapsto \mathbb{P}(\omega, F)$ is a version of $P(F|\mathcal{G})$;*

   i.e. fix F, $\mathbb{P}(\cdot, F) = Y(\cdot)$ is a random variable where Y is measurable to $\mathcal{G}$, $Y = E[I_F \mid \mathcal{G}]$ a.s..

   i.e. check $\forall G \in \mathcal{G}$,

   $$\int_G Y dP = \int_G I_F dP.$$

2. *for almost every ω, the section $F \mapsto \mathbb{P}(\omega, F)$ is a probability measure on $\mathcal{F}$.*

   i.e. $\exists N (\subseteq \Omega)$ with $P(N) = 0$ s.t. for (fixed)$\omega \in \Omega \setminus N$, $\mathbb{P}(\omega, \cdot) = \mu(\cdot)$ is a probability measure.

   i.e.

   (a) $\forall F \in \mathcal{F}$, $\mu(F) \geq \mu(\phi) = 0$ and $\mu(\Omega) = 1$.

   (b) if $(F_n) \in \mathcal{F}$ is a countable sequence of disjoint sets, then $\mu(\bigcup_{n=1}^{\infty} F_n) = \sum_{n=1}^{\infty} \mu(F_n)$.

Then such a map is called a regular conditional probability given $\mathcal{G}$.

Remark 5. *Let X be a random variables. We say $\mathbb{P}(\cdot, \cdot)$ is a regular conditional probability of X given $\mathcal{G}$ if the condition 1. holds for $F = \{\omega \in \Omega : X(\omega) \in A\}$ for $A \in \mathcal{B}(\mathbb{R})$, and condition 2. holds for $F_n = \{\omega \in \Omega : X(\omega) \in A_n\}$ for $A_1, A_2, \ldots$ are disjoint Borel sets.*

It's necessary to give the definition of regular conditional probability. There is an example to illustrate why we need to impose the condition 2.

Example 5. *Let $(F_n)_{n \geq 1}$, where each $F_n \in \mathcal{F}$, be a disjoint sequence.*

$$\begin{aligned} P\Big(\bigcup_{n=1}^{\infty} F_n \mid \mathcal{G}\Big) = E\Big[I_{\bigcup_{n=1}^{\infty} F_n} \mid \mathcal{G}\Big] &= E\Big[\sum_{n=1}^{\infty} I_{F_n} \mid \mathcal{G}\Big] \\ &= \sum_{n=1}^{\infty} E\Big[I_{F_n} \mid \mathcal{G}\Big] \quad \text{, by conditional MCT.} \\ &= \sum_{n=1}^{\infty} P\Big(F_n \mid \mathcal{G}\Big) \quad \text{a.s.} \end{aligned}$$

That is,

$$\textit{for each} \quad F^{(k)} := (F_n^{(k)})_{n\geq 1} \quad \exists N_k (\subseteq \Omega), N_k \in \mathcal{F}, \textit{where} P(N_k) = 0$$

$$\textit{s.t.} \quad P\Big(\bigcup_{n=1}^{\infty} F_n^{(k)} \mid \mathcal{G}\Big)(\omega) = \sum_{n=1}^{\infty} P\Big(F_n^{(k)} \mid \mathcal{G}\Big)(\omega), \quad \forall \omega \in \Omega \setminus N_k.$$

But there are uncountably many such sequences $F^{(k)}$ i.e. uncountably many null sets N_k, and we know that $P\Big(\bigcup_k N_k\Big) \neq 0$, since $\bigcup_k N_k$ may not in $\mathcal{F}$. That is, we cannot find a null set N such that $P(\cdot \mid \mathcal{G})(\omega)$ is a probability measure for each $\omega \in \Omega \setminus N$. So $P(\cdot \mid \mathcal{G})$ violates condition 2, it's not a regular conditional probability.

However, the elementary conditional density is a regular conditional probability, which we will use in a later section. Here is a simple example illustrate this.

Example 6. *Let X and Z be random variables map from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B})$. The elementary conditional density $f_{X|Z}(x \mid z) := \frac{f_{X,Z}(x,z)}{f_Z(z)}$,where $f_Z(z) > 0$, is a regular conditional density for X given $\mathcal{G}$, where $\mathcal{G} = \sigma(Z)$.*

Proof. Consider $F \in \mathcal{F}$ such that $F = \{\omega \in \Omega : X(\omega) = x \in A\} = \{X \in A\}$ for some $A \in \mathcal{B}$.
$\mathbb{P} : \Omega \times \mathcal{F} \to [0, 1]$ is defined by

$$\mathbb{P}(\omega, F) = \int I_A(x) f_{X|Z}(x \mid Z(\omega)) dx.$$

1. For a fixed set $F \in \mathcal{F}$ s.t. $F = \{X \in A\}$ for some $A \in \mathcal{B}$, with $Z(\omega) = z$.

$$\begin{aligned} \mathbb{P}(\omega, F) = Y(\omega) &= \int I_A(x) f_{X|Z}(x \mid Z(\omega)) dx \\ &= \int I_A(x) f_{X|Z}(x \mid z) dx \\ &= g(z) \quad \text{i.e. a function of z.} \end{aligned}$$

to check it is a version of $P(X \in A \mid Z) = E[I_A(X) \mid Z]$ a.s..

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i.e. To show this equality is valid

$$\int_G Y dP = \int_G I_A(X) dP, \quad \forall G \in \sigma(Z).$$

Write $G = \{\omega : Z(\omega) \in B\} = Z \in \mathcal{B}$, becomes

$$\int_{\{Z \in B\}} Y dP = \int_{\{Z \in B\}} I_A(X) dP.$$

$$\begin{aligned}
LHS &= \int g(z) I_B(z) f_Z(z) dz \quad \text{,where} g(z) = \int I_A(x) f_{X|Z}(x \mid z) dx. \\
&= \int \left(\int I_A(x) f_{X|Z}(x \mid z) dx \right) I_B(z) f_Z(z) dz \\
&= \int \int I_A(x) I_B(z) f_{X|Z}(x \mid z) f_Z(z) dx dz \quad \text{,by Fubini's thm.} \\
RHS &= \int \int I_A(x) I_B(z) f_{X,Z}(x, z) dx dz \\
&= \int \int I_A(x) I_B(z) f_{X|Z}(x \mid z) f_Z(z) dx dz
\end{aligned}$$

Therefore left hand side = right hand side.

2. For every $\omega \in \Omega$, $\mathbb{P}(\omega, F) = \mu(F) = \int I_A(x) f_{X|Z}(x \mid Z(\omega)) dx$, where $F = \{X \in A\}$ for some $A \in \mathcal{B}$ and $Z(\omega) = z$.

   (a) $\mu(\phi) = \int 0 \cdot f_{X|Z}(x \mid z) dx = 0$
   $\mu(\Omega) = \int 1 \cdot f_{X|Z}(x \mid z) dx = \int \frac{f_{X,Z}(x,z)}{f_Z(z)} dx = 1$

   (b) For countable disjoint sequence $(F_n)_{n \geq 1} \in \mathcal{F}$ s.t. $F_n = \{X \in A_n\}$ for some $A_n \in \mathcal{B}$, for each n.

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i.e. $(A_n)_{n\geq 1}$ is also disjoint.

$$\begin{aligned}\mu(\bigcup_{n=1}^{\infty} F_n) &= \int I_{\bigcup_{n=1}^{\infty} A_n}(x) f_{X|Z}(x \mid z)dx = \int \sum_{n=1}^{\infty} I_{A_n}(x) f_{X|Z}(x \mid z)dx \\ &= \sum_{n=1}^{\infty} \int I_{A_n}(x) f_{X|Z}(x \mid z)dx \quad \text{,by MCT} \\ &= \sum_{n=1}^{\infty} \mu(F_n)\end{aligned}$$

□

Example 7 (Recurrence)**.** *Suppose random variables X, Y, Z and T have a strictly positive joint density $f_{X,Y,Z,T}$ on $\mathbb{R}^4$. The recurrence property is illustrated by the expression :*

$$f_{T|X,Y,Z} = (f_{T|Z})_{|X,Y} := \frac{f_{T,Z|X,Y}}{f_{Z|X,Y}}.$$

Proof. Given that X, Y, Z, T follow a joint density $f_{X,Y,Z,T}$, we can determine $f_{X,Y,Z}$ as follows:

$$f_{X,Y,Z}(x, y, z) = \int_{\mathbb{R}} f_{X,Y,Z,T}(x, y, z, t)dt.$$

Similarly, we can compute $f_{X,Y}$ and other marginal densities.
This allows us to define

$$f_{T|X,Y,Z}(t|x, y, z) := \frac{f_{X,Y,Z,T}(x, y, z, t)}{f_{X,Y,Z}(x, y, z)}, \tag{3.5}$$

and

$$f_{T,Z|X,Y}(t, z|x, y) := \frac{f_{X,Y,Z,T}(x, y, z, t)}{f_{X,Y}(x, y)}, \tag{3.6}$$

also

$$f_{Z|X,Y}(z|x, y) := f_{X,Y,Z}(x, y, z)/f_{X,Y}(x, y). \tag{3.7}$$

All of these are regular conditional densities.

So

$$
\begin{aligned}
f_{T|X,Y,Z}(t|x,y,z) &= \frac{f_{X,Y,Z,T}(x,y,z,t)}{f_{X,Y,Z}(x,y,z)} \quad \text{, by equation (3.5)} \\
&= \frac{f_{X,Y,Z,T}(x,y,z,t)/f_{X,Y}(x,y)}{f_{X,Y,Z}(x,y,z)/f_{X,Y}(x,y)} \\
&= \frac{f_{T,Z|X,Y}(t,z|x,y)}{f_{Z|X,Y}(z|x,y)} \quad \text{, by equations (3.6) and (3.7)}
\end{aligned}
$$

Then define last equation by

$$(f_{T|Z})_{|X,Y} = \frac{f_{T,Z|X,Y}}{f_{Z|X,Y}}.$$

We have

$$f_{T|X,Y,Z} = (f_{T|Z})_{|X,Y} \text{ a recursive relation.}$$

□

Furthermore, the recursive relation can be deduced in more general case.

Property 3.3.1 (Recursive Relation)**.** *If $X, Y_1, \ldots, Y_n$ have a joint density, then the conditional density $f_{X|Y_1,\ldots,Y_n}$ can be expressed recursively as:*

$$f_{X|Y_1,\ldots,Y_n} = \left(f_{X|Y_n}\right)_{|Y_1,\ldots,Y_{n-1}}.$$

This property illustrates how the conditional density of X given multiple variables $Y_1, \ldots, Y_n$ can be decomposed into a recursive relationship involving the simpler conditional density $f_{X|Y_n}$ and the conditioning set $Y_1, \ldots, Y_{n-1}$.

3.3.1.2 Bayes's Formula for Bivariate Normal

Theorem 3.3.1 (Bayes theorem)**.** *Let X and Y be random variables with joint density f_{XY} on $\mathbb{R}^2$ and it's strictly positive. The formula*

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_X(x) f_{Y|X}(y|x)}{f_Y(y)}.$$

is called Bayes's formula.

As we know that $f_{X|Y}(x|y)$ is of parameter x and by the the fact that

$$\int_{\mathbb{R}} f_{X|Y}(x|y)dx = \int_{\mathbb{R}} \frac{f_{X,Y}(x,y)}{f_Y(y)}dx = \frac{\int_{\mathbb{R}} f_{X,Y}(x,y)dx}{f_Y(y)} = 1.$$

Then $f_Y(y)$ is computed by

$$f_Y(y) = \frac{\int_{\mathbb{R}} f_{X,Y}(x,y)dx}{\int_{\mathbb{R}} f_{X|Y}(x|y)dx}.$$

Thus it's a 'constant' of proportionality, depending on y, then we get a relation between the conditional density of $X|Y$ and that of $Y|X$

$$f_{X|Y}(x|y) \propto f_X(x) f_{Y|X}(y|x). \tag{3.8}$$

Lemma 3.3.1 (Bayes Lemma)**.** *Suppose that random variable X is distributed as $N(\mu, U)$ and the conditional distribution of $Y|X$ is distributed as $N(a + bX, W)$, where $\mu, a, b \in \mathbb{R}$ and numbers $U, V \in (0, \infty)$.*
Then the conditional distribution of $X|Y$ is distributed as $N(\hat{X}, V)$, where $V \in (0, \infty)$ and the random variable $\hat{X}$ are determined as

$$\frac{1}{V} = \frac{1}{U} + \frac{b^2}{W} \quad \textit{and} \quad \frac{\hat{X}}{V} = \frac{\mu}{U} + \frac{b(Y-a)}{W}$$

respectively.

Proof. X is distributed as $N(\mu, U)$ with density

$$f_X(x) = \frac{1}{\sqrt{2\pi U}} \exp\{\frac{-(x-\mu)^2}{2U}\}.$$

The conditional density of $Y|X$ is the density of $N(a + bX, W)$, that is,

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi W}} \exp\{\frac{-(y-(a+bx))^2}{2W}\}.$$

Using the relation (3.8),

$$\begin{aligned}
f_{X|Y}(x|y) &\propto \frac{1}{\sqrt{2\pi U}} \exp\{\frac{-(x-\mu)^2}{2U}\} \cdot \frac{1}{\sqrt{2\pi W}} \exp\{\frac{-(y-(a+bx))^2}{2W}\} \\
&\propto \exp\{\frac{-(x-\mu)^2}{2U}\} \cdot \exp\{\frac{-(y-(a+bx))^2}{2W}\} \\
&\propto \exp\{\frac{-(x-\mu)^2}{2U} + \frac{-(y-(a+bx))^2}{2W}\} \\
&\propto \exp\{-\frac{1}{2}\left[(\frac{1}{U}+\frac{b^2}{W})x^2 - 2(\frac{\mu}{U}+\frac{b(y-a)}{W})x\right]\} \\
&= \exp\{-\frac{1}{2}\left[\frac{1}{V}x^2 - \frac{2\hat{x}}{V}x\right]\}, \text{ where } \frac{1}{V} = \frac{1}{U}+\frac{b^2}{W} \text{ and } \frac{\hat{x}}{V} = \frac{\mu}{U}+\frac{b(y-a)}{W}. \\
&\propto \exp\{-\frac{1}{2}\left[\left(\frac{1}{\sqrt{V}}x - \frac{\hat{x}}{\sqrt{V}}\right)^2\right]\} \\
&= \exp\{-\frac{(x-\hat{x})^2}{2V}\}
\end{aligned}$$

Then it follows from that $X|Y=y \sim N(\hat{x}, V)$, where

$$\frac{1}{V} = \frac{1}{U}+\frac{b^2}{W} \quad \text{and} \quad \frac{\hat{x}}{V} = \frac{\mu}{U}+\frac{b(y-a)}{W}.$$

Thus $X|Y$ is distributed as $N(\hat{X}, V)$. □

Corollary 2. *Continuing the notation discussed in the previous lemma, we have*

$$\|X-\hat{X}\|_2^2 = E[(X-\hat{X})^2] = V.$$

Proof. Reminding that $\hat{X}$ is a function of Y.

Since $X|Y=y \sim N(\hat{x}, V)$, then

$$E[X|Y=y] = \hat{x} \quad \text{and} \quad Var[X|Y=y] := E[(X-\hat{x})^2|Y=y] = V.$$

Keeping Y random, we have

$$E[(X - \hat{X})^2|Y] = V \quad \text{a.s..}$$

Thus

$$E[(X - \hat{X})^2] = EE[(X - \hat{X})^2|Y] = E[V] = V.$$

□

3.3.2 Noisy Observation of a Single Variable

Consider a probability space $(\Omega, \mathcal{F}, P)$. Let X and $\eta_1, \eta_2, \ldots$ be independent random variables defined on $(\Omega, \mathcal{F}, P)$, where

$$X \sim N(0, \sigma^2) \quad \text{and} \quad (\eta_k)_{k\geq 1} \overset{\text{i.i.d.}}{\sim} N(0, 1),$$

with σ^2 being a finite positive number. Thus, X and each η_k belong to the space $\mathcal{L}^2$.

We define the sequence $\{Y_k\}_{k\geq 1}$ by

$$Y_k = X + c_k\eta_k, \quad \forall k \geq 1,$$

where $(c_k)_{k\geq 1}$ is a sequence of positive real numbers. Each Y_k represents a noisy observation of X.

Let $\mathcal{F}_n$ be the σ-algebra generated by $Y_1, Y_2, \ldots, Y_n$ for all $n \geq 1$, specifically:

$$\mathcal{F}_n = \sigma(Y_1, Y_2, \ldots, Y_n) \subset \mathcal{F}, \quad \forall n \geq 1.$$

The sequence $(\mathcal{F}_n)_{n\geq 1}$ forms an increasing filtration with respect to $\mathcal{F}$, capturing the growing information available as more observations Y_k are made.

We define the sequence $(M_n)_{n\geq 1}$ by

$$M_n := E[X \mid \mathcal{F}_n], \quad \forall n \geq 1,$$

which represents the conditional expectation of X given $\mathcal{F}_n$. It can be shown that (M_n) is $\mathcal{L}^2$-bounded. Specifically, since

$$M_n^2 = E[X \mid \mathcal{F}_n]^2 \leq E[X^2 \mid \mathcal{F}_n],$$

we have $E[M_n^2] \leq E[X^2] = \sigma^2$, which implies

$$\sup_{n \geq 1} E[M_n^2] \leq \sigma^2 < \infty.$$

According to Theorem 12.1 in David Williams's *Probability with Martingales* (1991, p. 111) [18]: Since $(M_n)_{n \geq 1}$ is $\mathcal{L}^2$-bounded, it follows that

$$M_n \to M_\infty \quad \text{almost surely (a.s.) and in } \mathcal{L}^2.$$

Moreover, the $\mathcal{L}^2$-boundedness implies that $(M_n)_{n \geq 1}$ is a uniformly integrable martingale.

By Lévy's upward theorem (Theorem 2.3.5), we also have

$$M_n \to E[X \mid \mathcal{F}_\infty] = M_\infty \quad \text{a.s. and in } \mathcal{L}^2, \quad \text{as } n \to \infty.$$

The sequence $M_n = E[X \mid \mathcal{F}_n]$ represents the best estimator of X given the available information in $\mathcal{F}_n$, in the least-squares sense. Similarly, $M_\infty = E[X \mid \mathcal{F}_\infty]$ represents the best possible estimator of X using all available information.

Given this, a natural and important question in the context of recursive estimation arises: under what conditions does the estimator $M_\infty = X$ a.s., which ensures consistency in the RLS framework? This leads us to the following theorem, which provides a precise condition for such consistency.

3.3.3 Consistency in RLS

Theorem 3.3.2 (Consistency of Recursive Estimation)**.** *Let X and $\eta_1, \eta_2, \ldots$ be independent random variables defined on a probability space $(\Omega, \mathcal{F}, P)$, where $X \sim N(0, \sigma^2)$ and $\eta_k \sim$*

$N(0,1)$ *for* $k \geq 1$. *Let* $\mathcal{F}_n = \sigma(Y_1, Y_2, \ldots, Y_n)$, *where* $Y_k = X + c_k\eta_k$ *represents noisy observations of* X, *with* c_k *being a sequence of positive real numbers. Define* $M_n = E[X \mid \mathcal{F}_n]$ *for each* $n \geq 1$.

Then:

1. *The sequence* M_n *converges almost surely and in* $\mathcal{L}^2$ *to* $M_\infty = E[X \mid \mathcal{F}_\infty]$ *as* $n \to \infty$, *ensuring the consistency of the recursive estimator.*

2. *Moreover,*

$$X = M_\infty \quad \textit{a.s.} \quad \textit{if and only if} \quad \sum_{k=1}^{\infty} \frac{1}{c_k^2} = \infty,$$

which establishes the necessary and sufficient condition for the estimator M_n *to consistently converge to the true value* X.

The consistency of RLS provides a theoretical foundation for the accuracy of the Kalman-Bucy filter. Since as more observation is processed, the estimate becomes more accurate and reliable.

Proof. Let $\mathcal{C}_n(\cdot)$ denote the regular conditional distribution given $Y_1, Y_2, \ldots, Y_n$, and assume $\mathcal{C}_0(X) = N(0, \sigma^2)$. Consider the sequence of observations given by

$$Y_n = X + c_n\eta_n,$$

where $c_n\eta_n \sim N(0, c_n^2)$. Therefore,

$$Y_n \mid X = x \sim N(x, c_n^2).$$

Since the noise variables η_n are independent, the sequence $(Y_n \mid X = x)_{n\geq 1}$ is independent as well.

We note that

$$Y_n \mid X = x \overset{\mathrm{d}}{=} Y_n \mid X = x, Y_1, \ldots, Y_{n-1},$$

which implies that the conditional distribution of Y_n given X and the past observations $Y_1, \ldots, Y_{n-1}$ is the same as that given only X. Hence,

$$\mathcal{C}_{n-1}(Y_n \mid X) = N(X, c_n^2). \tag{3.9}$$

Where $\mathcal{C}_{n-1}(Y_n|X)$ stands for the regular conditional distribution of $Y_n \mid X$ given $Y_1, Y_2, \ldots, Y_{n-1}$.

Assume that

$$\mathcal{C}_{n-1}(X) = N(\hat{X}_{n-1}, V_{n-1}), \tag{3.10}$$

where $\hat{X}_{n-1}$ is a linear function of $Y_1, \ldots, Y_{n-1}$ and $V_{n-1} \in (0, \infty)$ is a constant.

Combining equations (3.9) and (3.10), and applying the Bayes lemma (Lemma 3.3.1), we obtain

$$\mathcal{C}_{n-1}(X \mid Y_n) = N(\hat{X}_n, V_n),$$

where

$$\frac{1}{V_n} = \frac{1}{V_{n-1}} + \frac{1}{c_n^2} \quad \text{and} \quad \frac{\hat{X}_n}{V_n} = \frac{\hat{X}_{n-1}}{V_{n-1}} + \frac{Y_n}{c_n^2}.$$

Here, $\mathcal{C}_{n-1}$ is a regular conditional probability, allowing us to apply the Bayes lemma.

By the recursive relation , property 3.3.1, we have

$$\mathcal{C}_n(X) = \mathcal{C}_{n-1}(X \mid Y_n) = N(\hat{X}_n, V_n),$$

and thus,

$$X \mid Y_1, \ldots, Y_n \sim N(\hat{X}_n, V_n).$$

Recall that $M_n = E[X \mid \mathcal{F}_n]$, so $M_n = \hat{X}_n$.

From the discussion above, M_∞ is the $\mathcal{L}^2$-limit of (M_n). If X is also the $\mathcal{L}^2$-limit of (M_n), then by the uniqueness of the $\mathcal{L}^2$ limit, $X = M_\infty$ almost surely.

It follows immediately that

$$\begin{aligned} E[(X - M_n)^2] &= E[(X - \hat{X}_n)^2] = V_n \\ &= \frac{1}{\frac{1}{\sigma^2} + \sum_{k=1}^{n} \frac{1}{c_k^2}}. \end{aligned} \tag{*}$$

Therefore,

$$\lim_{n\to\infty} E[(X - M_n)^2] = \lim_{n\to\infty} \frac{1}{\frac{1}{\sigma^2} + \sum_{k=1}^{n} \frac{1}{c_k^2}} = 0,$$

which implies that X is the $\mathcal{L}^2$ limit of M_n.

This holds if and only if

$$\sum_{k=1}^{\infty} \frac{1}{c_k^2} = \infty.$$

To compute (*), we observe:

$$\begin{aligned}
\frac{1}{V_n} &= \frac{1}{V_{n-1}} + \frac{1}{c_n^2} \\
&= \left(\frac{1}{V_{n-2}} + \frac{1}{c_{n-1}^2} \right) + \frac{1}{c_n^2} \\
&= \left(\frac{1}{V_{n-3}} + \frac{1}{c_{n-2}^2} \right) + \frac{1}{c_{n-1}^2} + \frac{1}{c_n^2} \\
&\vdots \\
&= \frac{1}{V_0} + \frac{1}{c_1^2} + \sum_{k=2}^{n} \frac{1}{c_k^2}.
\end{aligned}$$

Given that $\mathcal{C}_0(X) = N(0, \sigma^2)$, we have $V_0 = \sigma^2$, so

$$V_n = \frac{1}{\frac{1}{\sigma^2} + \sum_{k=1}^{n} \frac{1}{c_k^2}}.$$

□

The recursive estimation process derived here is directly analogous to the Kalman filter, where similar updates are used to estimate the state of a system from noisy observations.

In the next section, the Kalman filter will be introduced and show how it extends the procedure discussed in this proof to dynamic systems.

3.4 Kalman filter

Both the Kalman-Bucy filter and RLS are based on recursive estimation principles. While the Kalman-Bucy filter extends these ideas to dynamic systems, where the state evolves over time.

This thesis introduces the simplest case: a linear stochastic state-space model in which all matrices and random vectors reduce to 1-dimensional forms. This simplification allows us to

clearly illustrate the fundamental principles and workings of the Kalman filter without the added complexity of higher dimensions.

Then the dynamic system can be modelled by

Definition 3.4.1. *Let A_n, H_n, C_n, K_n, u_n be real values.*
the state equation:

$$X_{n+1} = A_n X_n + H_n \epsilon_n + u_n \tag{3.11}$$

and the observation equation:

$$Y_n = C_n X_n + K_n \eta_n, \quad \forall n \geq 0 \tag{3.12}$$

Here X_n is the state of the system. Y_n is the output of the system, and u_n is an input value of the system.
Let $(\epsilon_n)_{n\geq 0}$ and $(\eta_n)_{n\geq 0}$ be iid sequences of $N(0,1)$.

Let the initial state $X_0 \sim N(0, \sigma^2)$, and let $Y_0 = 0$ signify that no observations have been obtained yet. Also assume that $X_0, Y_0, \epsilon_k, \eta_l$ for $k, l \geq 1$ are independent.

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3.4.1 Overview of the Calculation

As mentioned at first, Kalman filter aims to produce the best estimation of state X_n given available information up to time n at each time. Instead of doing directly, Kalman filter carry out through a two-step procedure: predicting and correction. That is, the dynamics of Kalman filter results from a recursive application of a prediction step and a correction step.

Noted that 'available information up to time n' is comprised of observations up to time n, $(Y_1, \ldots, Y_n)$, and the input value $(u_0, \ldots, u_{n-1})$. For convenience, the notation or the statement in the following will omit the input values

Denote $\mathcal{C}_n(\cdot)$ by the regular conditional distribution of given $Y_1, \ldots, Y_n$. So the estimator produced by Kalman filter for each state X_n is written by $\mathcal{C}_n(X_n)$.

Let's take a quick look at how the Kalman filter works. Start from X_0, we input the control value u_0. Before obtaining the observation, we make a prediction $\mathcal{C}_0(X_1)$ only based on u_0. This is the **"prediction step"**, and $\mathcal{C}_0(X_1)$ is called a **prediction estimator** of X_1.

Next, when the observation Y_1 is available, we improve the prediction $\mathcal{C}_0(X_1)$ to be $\mathcal{C}_1(X_1)$. It's the **"correction step"** or the **"update step"**.

Combining these two steps, we get the estimator $\mathcal{C}_1(X_1)$ at time 1 from the filter. Repeating the procedures, get the prediction $\mathcal{C}_1(X_2)$ and then correct it to get $\mathcal{C}_2(X_2)$. Hence we obtain a sequence of **filtering estimators** $\mathcal{C}_n(X_n)$ for $n \geq 1$ from the Kalman filter.

3.4.2 Derivations

3.4.2.1 Independent and Normal Properties

In the following discussions, we further assume that the system is without input values. That is, $u_0 = u_1 = \cdots = 0$ in equations (3.11) and (3.12) and this allows us to focus solely on the estimation problem.

Rewrite equations (3.11) and (3.12),

$$\begin{aligned} X_{n+1} &= A_n X_n + H_n \epsilon_n \\ &= A_n \left(A_{n-1} X_{n-1} + H_{n-1} \epsilon_{n-1} \right) + H_n \epsilon_n \\ &= A_n A_{n-1} X_{n-1} + A_n H_{n-1} \epsilon_{n-1} + H_n \epsilon_n \\ &\vdots \\ &= \left(\prod_{k=0}^{n} A_{n-k} \right) X_0 + \sum_{k=0}^{n} \left(\prod_{j=0}^{k-1} A_{n-j} \right) H_{n-k} \epsilon_{n-k} \end{aligned} \tag{3.13}$$

$$
\begin{aligned}
Y_n &= C_n X_n + K_n \eta_n \\
&= C_n \left(\prod_{k=0}^{n-1} A_{n-1-k} \right) X_0 + C_n \sum_{k=0}^{n-1} \left(\prod_{j=0}^{k-1} A_{n-1-j} \right) H_{n-1-k} \epsilon_{n-1-k} + K_n \eta_n
\end{aligned} \tag{3.14}
$$

From the above computation, equations (3.13) and (3.14), it follows that all considered random variables are centered. Specifically, since the noise terms ϵ_n and η_n are assumed to be centered (i.e., they have a mean of zero), the state X_n and the observation Y_n, being linear combinations of these noise terms and the initial state X_0, are also centered.

In Kalman filter setting $X_0, Y_0, \epsilon_0, \epsilon_1 \ldots, \eta_1, \eta_2, \ldots$ are independent, thus they have joint normal distribution. So the condition in Theorem A.1.3 is satisfied, a closed linear span

$$
G := \overline{span}\{X_0, Y_0, \epsilon_0, \epsilon_1 \ldots, \eta_1, \eta_2, \ldots\}
$$

is a Gaussian space.

Property 3.4.1. *$\{(X_n)_{n\geq 0}, (Y_m)_{m\geq 0}, (\epsilon_k)_{k\geq 0}, (\eta_l)_{l\geq 1}\}$ have a joint normal distribution.*

Property 3.4.2. *For every $n \geq 0$, ϵ_n is independent with $X_n, X_{n-1}, \ldots, X_0$, and ϵ_n is independent with $Y_n, Y_{n-1}, \ldots, Y_0$.*

Property 3.4.3. *$(\eta_l)_{l\geq 1}$ is independent with $(X_n)_{n\geq 0}$, and for every $l \geq 1$, η_l is independent of Y_m for $m \neq l$.*

Proof. 3.4.1. We have $X_n \in G$ for all $n \geq 0$ and $Y_m \in G$ for all $m \geq 0$. Since they are linear combinations (from (3.13) (3.14)) of elements in G. By Theorem A.1.1 any set in Gaussian space has joint normal distribution. □

Proof. 3.4.2. From equation (3.13), for every n, X_n depends on X_0 and $\epsilon_0, \ldots, \epsilon_{n-1}$. For a fixed n, ϵ_n is independent of $X_0, \epsilon_0, \ldots, \epsilon_{n-1}$, thus it's independent of X_n.
Similarly, for every n, Y_n depends on X_n and η_n. And for a fixed n, ϵ_n is independent of X_n, η_n, thus is independent of Y_n. □

Proof. 3.4.3. For every $l \geq 1$, η_l is independent of X_0 and $\epsilon_0, \ldots, \epsilon_n \quad \forall n \geq 0$ (and therefore every X_{n+1}), thus it's independent of $(X_n)_{n\geq 0}$. Then $(\eta_l)_{l\geq 1}$ is independent of $(X_n)_{n\geq 0}$.
We have already known that $\{\eta_l\}_{l\geq 1}$ is independent of $\{X_n\}_{n\geq 0}$. Each Y_m is defined as: $Y_m = CX_m + K\eta_m$ Given that η_l is independent of η_m for $m \neq l$ and η_l is independent of X_m for all m, it follows that η_l is independent of Y_m for $m \neq l$. Thus, η_l is independent of Y_m for $m \neq l$. □

From property 3.4.1 and Theorem A.1.1, $(Y_n)_{n\geq 0}$ is joint normal. Thus $(Y_n)_{n\geq 0}$ also generate a Gaussian space.

By Theorem A.1.4, for all $n \geq 0$, $X_n|Y_1, \ldots, Y_k$ is of normal distribution, for all k. Let $\hat{X}_{n|k}$ denote the conditional mean of $X_n|Y_1, \ldots, Y_k$ and $V_{n|k}$ denote the conditional variance.

Then write

$$\mathcal{C}_{n-1}(X_{n-1}) = N(\hat{X}_{n-1|n-1}, V_{n-1|n-1}).$$

3.4.2.2 Prediction step:

Find $\mathcal{C}_{n-1}(X_n) = N(\hat{X}_{n|n-1}, V_{n|n-1})$ before we obtain observation Y_n. Using only the information available up to time $n-1$ to make an estimator of state n.

From the state equation (3.11): $X_n = A_{n-1}X_{n-1} + H_{n-1}\epsilon_{n-1}$, get

$$\begin{aligned}\mathcal{C}_{n-1}(X_n) &= N(\hat{X}_{n|n-1}, V_{n|n-1}) \\ &= N(A_{n-1}\hat{X}_{n-1|n-1}, A_{n-1}^2 V_{n-1|n-1} + H_{n-1}^2)\end{aligned} \tag{3.15}$$

Since $\mathcal{C}_{n-1}(A_{n-1}X_{n-1}) = N(A_{n-1}\hat{X}_{n-1}, A_{n-1}^2 V_{n-1})$, and by property 3.4.2, ϵ_n is independent of $Y_1, \ldots, Y_{n-1}$ then $\mathcal{C}_{n-1}(H_n\epsilon_n) = N(0, H^2)$. By property 3.4.2 again we have the independence of X_{n-1} and ϵ_{n-1} thus the linearity of normal is valid.

3.4.2.3 Correction step:

Find $\mathcal{C}_n(X_n) = N(\hat{X}_{n|n}, V_{n|n})$ when the observation Y_n is available. Make a correction of prediction estimator by taking Y_n into consideration.

From the observation equation (3.12): $Y_n = C_n X_n + K_n \eta_n$, giving $Y_1, \ldots, Y_{n-1}$ and X_n, we get

$$\mathcal{C}_{n-1}(Y_n|X_n) = N(C_n X_n, K_n^2). \tag{3.16}$$

Combine (3.15) and (3.16) by Bayse Lemma (Lemma 3.3.1) and the recursive relation (property 3.3.1) , we get

$$\mathcal{C}_{n-1}(X_n|Y_n) = \mathcal{C}_n(X_n) = N(\hat{X}_{n|n}, V_{n|n})$$

, where

$$\frac{1}{V_{n|n}} = \frac{1}{V_{n|n-1}} + \frac{C_n^2}{K_n^2} \tag{3.17}$$

and

$$\frac{\hat{X}_{n|n}}{V_{n|n}} = \frac{\hat{X}_{n|n-1}}{V_{n|n-1}} + \frac{C_n Y_n}{K^2}. \tag{3.18}$$

To get $V_{n|n}$ from equation (3.17),

$$\begin{aligned} V_{n|n} &= \left(\frac{1}{V_{n|n-1}} + \frac{C_n^2}{K_n^2}\right)^{-1} \\ &= V_{n|n-1} - \frac{C_n^2 V_{n|n-1}^2}{K_n^2 + C_n^2 V_{n|n-1}} \\ &= \left(1 - \frac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}} \cdot C_n\right) V_{n|n-1} \\ &= (1 - \mathbb{K}_n C_n) V_{n|n-1} \quad \square \end{aligned}$$

, where $\mathbb{K}_n := \dfrac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}}$ is the Kalman gain, and the property of it will be introduced later.

To get $\hat{X}_{n|n}$ from equation (3.18),

$$
\begin{aligned}
\hat{X}_{n|n} &= V_{n|n} \cdot \frac{\hat{X}_{n|n}}{V_{n|n}} \\
&= (1 - \mathbb{K}_n C_n) V_{n|n-1} \cdot \left(\frac{\hat{X}_{n|n-1}}{V_{n|n-1}} + \frac{C_n Y_n}{K_n^2} \right) \\
&= (1 - \mathbb{K}_n C_n) \hat{X}_{n|n-1} + (1 - \mathbb{K}_n C_n) V_{n|n-1} \frac{C_n Y_n}{K_n^2} \\
&= \hat{X}_{n|n-1} - \mathbb{K}_n C_n \hat{X}_{n|n-1} + \frac{C_n V_{n|n-1}}{K_n^2} \cdot Y_n - \frac{\mathbb{K}_n C_n^2 V_{n|n-1}}{K_n^2} \cdot Y_n \\
&= \hat{X}_{n|n-1} + \left(\frac{C_n V_{n|n-1}}{K_n^2} - \frac{\mathbb{K}_n C_n^2 V_{n|n-1}}{K_n^2} \right) Y_n - \mathbb{K}_n C_n \hat{X}_{n|n-1} \\
&= \hat{X}_{n|n-1} + \left(\mathbb{K}_n (1 + \frac{C_n^2 V_{n|n-1}}{K_n^2}) - \frac{\mathbb{K}_n C_n^2 V_{n|n-1}}{K_n^2} \right) Y_n - \mathbb{K}_n C_n \hat{X}_{n|n-1} \\
&= \hat{X}_{n|n-1} + \left(\mathbb{K}_n + \frac{\mathbb{K}_n C_n^2 V_{n|n-1}}{K_n^2} - \frac{\mathbb{K}_n C_n^2 V_{n|n-1}}{K_n^2} \right) Y_n - \mathbb{K}_n C_n \hat{X}_{n|n-1} \\
&= \hat{X}_{n|n-1} + \mathbb{K}_n Y_n - \mathbb{K}_n C_n \hat{X}_{n|n-1} \\
&= \hat{X}_{n|n-1} + \mathbb{K}_n (Y_n - C_n \hat{X}_{n|n-1}) \quad \square
\end{aligned}
$$

And it states how to correct the prediction estimator after getting the latest observation Y_n.

The derivation above for finding $\hat{X}_{n|n}$ and $V_{n|n}$ is referred to the matrix version in [14].

3.4.3 Summary of key equations

Since the conditional densities propagated by the Kalman filter are all Gaussian, that is characterized by mean and variance. So we just focus on the dynamics of mean and variance of the conditional distribution instead of discussing the entire density function.

Model:

$$X_{n+1} = A_n X_n + H_n \epsilon_n$$

$$Y_n = C_n X_n + K_n \eta_n, \quad \forall n \geq 1$$

where A_n, H_n, C_n, K_n are deterministic real values, and $(\epsilon_n)_{n \geq 0}$, $(\eta_n)_{n \geq 1}$ are iid sequences of

$N(0,1)$. $X_0 \sim N(0,\sigma^2), Y_0 = 0$ and $X_0, Y_0, \epsilon_0, \epsilon_1, \ldots, \eta_l, \eta_2, \ldots$ are independent.

1. **Prediction step**

(Prediction mean) $\hat{X}_{n|n-1} = A_{n-1}\hat{X}_{n-1|n-1}$ (3.19)

(Prediction variance) $V_{n|n-1} = A_{n-1}^2 V_{n-1|n-1} + H_{n-1}^2$ (3.20)

2. **Correction step**

(Filtering mean) $\hat{X}_{n|n} = \hat{X}_{n|n-1} + \mathbb{K}_n(Y_n - C_n\hat{X}_{n|n-1})$ (3.21)

(Filtering variance) $V_{n|n} = (1 - \mathbb{K}_n C_n)V_{n|n-1}$ (3.22)

(Kalman gain) $\mathbb{K}_n = \dfrac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}}$ (3.23)

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3.4.4 The Best Linear Filter: minimize the mean-squared error

To measure how well the estimators perform, the errors will be defined.

- (prediction error) $\tilde{X}_{n|n-1} := X_n - \hat{X}_{n|n-1}$.
- (filtering error) $\tilde{X}_{n|n} := X_n - \hat{X}_{n|n}$.

The derivation done earlier, Section 3.4.2, is based on the principles of RLS with a Bayesian approach, Section 3.3.3. For the system (3.11) and (3.12), the filtering estimator $\hat{X}_{n|n}$, derived through the Kalman filter, achieves the minimum mean-squared error. That is, $E[(X_n - \hat{X}_{n|n})^2]$

is minimized.

To further verify this optimality, we explore the geometric interpretation of conditional expectation. This verification will not only show that Kalman filter's estimate is indeed the best in the least-squares sense, but also bridge the connection between the Bayesian approach and the geometrical insights into conditional expectation.

Remark 6. *The filtering mean-squared error is written by* $E(X_n - \hat{X}_{n|n})^2$.
We have the distribution that $\mathcal{C}_n(X_n) = N(\hat{X}_{n|n}, V_{n|n})$ *from Corollary 2,* $E(X_n - \hat{X}_{n|n})^2 = \|X_n - \hat{X}_{n|n}\|_2^2 = V_{n|n}$.

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3.4.4.1 Geometry and Projection

Consider the Gaussian space we defined before

$$G = \overline{span}\{X_0, Y_0, \epsilon_0, \epsilon_1 \ldots, \eta_1, \eta_2, \ldots\}.$$

as well as the σ-algebra generated by the observations $Y_1, Y_2, \ldots, Y_k$,

$$\mathcal{F}_k = \sigma(Y_1, Y_2, \ldots, Y_k).$$

Define $\mathcal{L}(Y_1^k)$ as the linear subspace spanned by the observations up to time k

$$\mathcal{L}(Y_1^k) := span\{Y_1, \ldots, Y_k\} = \{\sum_{i=1}^{k} a_i Y_i : a_i \in \mathbb{R}\}.$$

By property 3.4.1 and Theorem A.1.1 , the sequence $(Y_n)_{n\geq 0}$ is joint normal. Consequently, any finite subset of $(Y_n)_{n\geq 0}$ is also joint normal. Therefore by Theorem A.1.2 the space $\mathcal{L}(Y_1^k) \subseteq G$ is a Gaussian space for all $k \geq 0$.

Now, consider the filtering estimator of state n, reminding that $\hat{X}_{n|n}$ is the mean of $\mathcal{C}_n(X_n)$.

Therefore,

$$\hat{X}_{n|n} = E[X_n|\mathcal{F}_n].$$

By Theorem A.1.4, we know that $\hat{X}_{n|n}$ is the orthogonal projection of X_n onto space $\mathcal{L}(Y_1^n)$. Therefore $\hat{X}_{n|n} = \sum_{i=1}^{n} a_i Y_i$ for some coefficients $a_i \in \mathbb{R}, 1 \leq i \leq n$. That is, $\hat{X}_{n|n}$ is a linear estimator based on the observations.

Similarly, the prediction estimator $\hat{X}_{n|n-1}$ is the orthogonal projection of X_n onto the space $\mathcal{L}(Y_1^{n-1})$.

Additionally, we define the following notations related to the observations:

- Let $\hat{Y}_{n|n-1}$ be the orthogonal projection of Y_n onto space $\mathcal{L}(Y_1^{n-1})$.
  In detail, we have

$$\begin{aligned}
\hat{Y}_{n|n-1} &= E[Y_n|Y_1, \ldots, Y_{n-1}] \\
&= E[C_n X_n + K\eta_n|Y_1, \ldots, Y_{n-1}] \quad \text{by observation (3.12).} \\
&= C_n E[X_n|Y_1, \ldots, Y_{n-1}] + K_n E[\eta_n] \quad \text{by property 3.4.3.} \\
&= C_n \hat{X}_{n|n-1} \quad \square
\end{aligned}$$

- Define the observation prediction error as $\tilde{Y}_{n|n-1} := Y_n - \hat{Y}_{n|n-1}$. Also write

$$\mathcal{L}(\tilde{Y}_{n|n-1}) := \{b\tilde{Y}_{n|n-1} : b \in \mathbb{R}\}$$

  for the linear subspace spanned by $\tilde{Y}_{n|n-1}$. Then $\mathcal{L}(\tilde{Y}_{n|n-1})$ is the orthogonal complement of $\mathcal{L}(Y_1^{n-1})$. In other words, the elements in $\mathcal{L}(\tilde{Y}_{n|n-1})$ are orthogonal to the elements in $\mathcal{L}(Y_1^{n-1})$.

Property 3.4.4. *The subspaces $\mathcal{L}(Y_1^{n-1})$ and $\mathcal{L}(\tilde{Y}_{n|n-1})$ are linear subspaces of $\mathcal{L}(Y_1^n)$, such that $\mathcal{L}(Y_1^n)$ is the direct sum of $\mathcal{L}(Y_1^{n-1})$ and $\mathcal{L}(\tilde{Y}_{n|n-1})$. That is,*

$$\mathcal{L}(Y_1^n) = \mathcal{L}(Y_1^{n-1}) \oplus \mathcal{L}(\tilde{Y}_{n|n-1}).$$

Recall from equation (3.21) that the expression for the filtering estimator of the state at time n can be rewritten in terms of the observation prediction error:

$$\begin{aligned}\hat{X}_{n|n} &= \hat{X}_{n|n-1} + \mathbb{K}_n(Y_n - C\hat{X}_{n|n-1}) \\ &= \hat{X}_{n|n-1} + \mathbb{K}_n\tilde{Y}_{n|n-1}\end{aligned} \tag{3.24}$$

where $\hat{X}_{n|n-1} \in \mathcal{L}(Y_1^{n-1})$ and $\tilde{Y}_{n|n-1} \in \mathcal{L}(\tilde{Y}_{n|n-1})$ are orthogonal. This gives us an orthogonal decomposition of $\hat{X}_{n|n}$.

Based on the above content, we obtain the following figure.

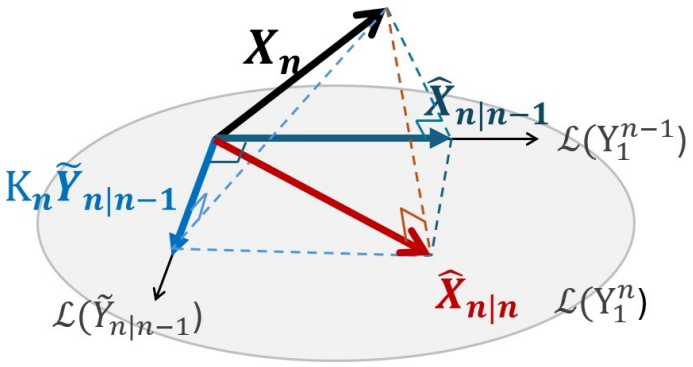


Figure 3.1: Kalman Filter Projection and Orthogonality

where

- $\hat{X}_{n|n}$ is the orthogonal projection of X_n onto space $\mathcal{L}(Y_1^n)$.
- $\hat{X}_{n|n-1}$ is the orthogonal projection of X_n onto space $\mathcal{L}(Y_1^{n-1})$.
- $\hat{X}_{n|n} = \hat{X}_{n|n-1} + \mathbb{K}_n\tilde{Y}_{n|n-1}$ is an orthogonal decomposition.

Notice from Figure 3.1, $\mathbb{K}_n\tilde{Y}_{n|n-1}$ can also be viewed as the orthogonal projection of X_n onto $\mathcal{L}(\tilde{Y}_{n|n-1})$. Then $\mathbb{K}_n$ is the coefficient that minimizes $E[(X_n - \mathbb{K}_n\tilde{Y}_{n|n-1})^2]$.

Also, according to the proposition of projection coefficient, a standard result in linear algebra, $\mathbb{K}_n$ can be expressed as

$$\mathbb{K}_n = \frac{E[X_n \tilde{Y}_{n|n-1}]}{E[\tilde{Y}^2_{n|n-1}]} = \frac{C_n V_{n|n-1}}{C_n^2 V_{n|n-1} + K_n^2}. \tag{3.25}$$

This is consistent with equation (3.23), the result derived in the correction step. The result and details can also be found in section 3.1 of Davis's *Stochastic Modelling and Control* (2013, p.102) [10].

Proof. Express X_n as the sum of the filtering error and the prediction estimator i.e. $X_n = \tilde{X}_{n|n-1} + \hat{X}_{n|n-1}$. This gives us an orthogonal decomposition of X_n.

Compute numerator:

$$\begin{aligned}
E[X_n \tilde{Y}_{n|n-1}] &= E[X_n(C_n \tilde{X}_{n|n-1} + K_n \eta_n)] \\
&= C_n E[X_n \tilde{X}_{n|n-1}] + K_n E[\eta_n] \quad \text{, where } E[\eta_n] = 0 \\
&= C_n E[(\hat{X}_{n|n-1} + \tilde{X}_{n|n-1})\tilde{X}_{n|n-1}] \\
&= C_n E[\tilde{X}^2_{n|n-1}] \\
&= C_n V_{n|n-1}
\end{aligned}$$

Compute denominator:

$$\begin{aligned}
E[\tilde{Y}^2_{n|n-1}] &= E[(C_n \tilde{X}_{n|n-1} + K_n \eta_n)^2] \\
&= E[C_n^2 \tilde{X}^2_{n|n-1} + 2C_n \tilde{X}_{n|n-1} K_n \eta_n + K_n^2 \eta_n^2] \\
&\quad \text{, where } \tilde{X}_{n|n-1} \text{ is independent with } \eta_n \\
&= C_n^2 E[\tilde{X}^2_{n|n-1}] + K_n^2 E[\eta_n^2] \\
&= C_n^2 V_{n|n-1} + K_n^2
\end{aligned}$$

Thus we get the equation (3.25). □

Claim 1. *We claim that the Kalman gain* $\mathbb{K}_n = \dfrac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}}$ *is also the coefficient such that* $E[(X_n - \hat{X}_{n|n})^2]$ *is minimized.*

In other words, $\mathbb{K}_n$ achieves the minimal mean-squared error between the system state and the filtering estimator. This claim will be verified below using the orthogonality derived earlier, leveraging the geometric interpretation of the Kalman filter.

Proof. Which can be seen directly from the figure 3.1.

$$
\begin{aligned}
E[(X_n - \hat{X}_{n|n})^2] &= E\big[\big(X_n - (\hat{X}_{n|n-1} + \mathbb{K}_n \tilde{Y}_{n|n-1})\big)^2\big] \quad \text{by (3.24)} \\
&= E\big[\big((X_n - \mathbb{K}_n \tilde{Y}_{n|n-1}) - \hat{X}_{n|n-1}\big)^2\big] \\
&= E\big[(X_n - \mathbb{K}_n \tilde{Y}_{n|n-1})^2\big] + 2E\big[(X_n - \mathbb{K}_n \tilde{Y}_{n|n-1})\hat{X}_{n|n-1}\big] + E\big[\hat{X}^2_{n|n-1}\big] \\
&= E\big[(X_n - \mathbb{K}_n \tilde{Y}_{n|n-1})^2\big] + 2E\big[X_n \hat{X}_{n|n-1}\big] - 2E\big[\mathbb{K}_n \tilde{Y}_{n|n-1}\hat{X}_{n|n-1}\big] + E\big[\hat{X}^2_{n|n-1}\big]
\end{aligned}
$$

Since X_n is orthogonal to $\hat{X}_{n|n-1}$ then $E\big[X_n \hat{X}_{n|n-1}\big] = 0$. And $\mathbb{K}_n \tilde{Y}_{n|n-1}$ is orthogonal to $\hat{X}_{n|n-1} (\in \mathcal{L}(Y_1^{n-1}))$ then $E\big[\mathbb{K}_n \tilde{Y}_{n|n-1}\hat{X}_{n|n-1}\big] = 0$.

Hence

$$
E[(X_n - \hat{X}_{n|n})^2] = E\big[(X_n - \mathbb{K}_n \tilde{Y}_{n|n-1})^2\big] E\big[\hat{X}^2_{n|n-1}\big].
$$

Since $\mathbb{K}_n$ minimizes $E\big[(X_n - \mathbb{K}_n \tilde{Y}_{n|n-1})^2\big]$, it consequently also minimizes $E[(X_n - \hat{X}_{n|n})^2]$. □

3.4.4.2 Another Derivation of Kalman Gain $\mathbb{K}_n$

We can obtain the Kalman gain $\mathbb{K}_n$ by minimizing the expected squared error $E[(X_n - \hat{X}_{n|n})^2]$ using calculus techniques. This is a common approach in most materials.

First, express the filtering error:

$$
\begin{aligned}
X_n - \hat{X}_{n|n} &= X_n - \Big(\hat{X}_{n|n-1} + \mathbb{K}_n (Y_n - C_n \hat{X}_{n|n-1})\Big) \quad \text{by equation (3.21)} \\
&= X_n - \hat{X}_{n|n-1} - \mathbb{K}_n (C_n X_n + K_n \eta_n - C_n \hat{X}_{n|n-1}) \quad \text{by equation (3.12)} \\
&= X_n - \hat{X}_{n|n-1} - \mathbb{K}_n (C_n X_n + K_n \eta_n) + \mathbb{K}_n C_n \hat{X}_{n|n-1} \\
&= (1 - \mathbb{K}_n C_n) X_n - (1 - \mathbb{K}_n C_n) \hat{X}_{n|n-1} - \mathbb{K}_n K_n \eta_n \\
&= (1 - \mathbb{K}_n C_n)(X_n - \hat{X}_{n|n-1}) - \mathbb{K}_n K_n \eta_n.
\end{aligned}
$$

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Next, we minimize the expected squared error:

$$
\begin{aligned}
E[(X_n - \hat{X}_{n|n})^2] &= E\left[(1 - \mathbb{K}_n C_n)(X_n - \hat{X}_{n|n-1}) - \mathbb{K}_n K_n \eta_n\right]^2 \\
&= E\left[(1 - \mathbb{K}_n C_n)^2 (X_n - \hat{X}_{n|n-1})^2 + \mathbb{K}_n^2 K_n^2 \eta_n^2\right] \\
&\quad + 2(1 - \mathbb{K}_n C_n)\mathbb{K}_n K_n E\left[(X_n - \hat{X}_{n|n-1})\eta_n\right] \\
&= (1 - \mathbb{K}_n C_n)^2 E\left[(X_n - \hat{X}_{n|n-1})^2\right] + \mathbb{K}_n^2 K_n^2 E\left[\eta_n^2\right] \\
&\quad + 2(1 - \mathbb{K}_n C_n)\mathbb{K}_n K_n E\left[(X_n - \hat{X}_{n|n-1})\right] E\left[\eta_n\right] \\
&\qquad \text{, where } (X_n - \hat{X}_{n|n-1}) \text{ is independent of } \eta_n. \\
&\qquad \text{(for } X_0, \epsilon_0, \epsilon_1 \ldots, \eta_1, \eta_2, \ldots \text{ are independent.)} \\
&= (1 - 2\mathbb{K}_n C_n + \mathbb{K}_n^2 C_n^2) E\left[(X_n - \hat{X}_{n|n-1})^2\right] + \mathbb{K}_n^2 K_n^2 + 0 \quad \text{, since } \eta_n \sim N(0,1).
\end{aligned}
$$

To find the optimal $\mathbb{K}_n$, we set the derivative of the expected squared error with respect to $\mathbb{K}_n$ to zero:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\mathbb{K}_n} E[(X_n - \hat{X}_{n|n})^2] &= \frac{\mathrm{d}}{\mathrm{d}\mathbb{K}_n}\left((1 - 2\mathbb{K}_n C_n + \mathbb{K}_n^2 C_n^2) E[(X_n - \hat{X}_{n|n-1})^2] + \mathbb{K}_n^2 K_n^2\right) \\
&= (-2C_n + 2\mathbb{K}_n C_n^2) E[(X_n - \hat{X}_{n|n-1})^2] + 2\mathbb{K}_n K_n^2 = 0.
\end{aligned}
$$

Solving for $\mathbb{K}_n$:

$$
\begin{aligned}
C_n E[(X_n - \hat{X}_{n|n-1})^2] &= C_n^2 \mathbb{K}_n E[(X_n - \hat{X}_{n|n-1})^2] + \mathbb{K}_n K_n^2 \\
(K_n^2 + C_n^2 E[(X_n - \hat{X}_{n|n-1})^2])\mathbb{K}_n &= C_n E[(X_n - \hat{X}_{n|n-1})^2].
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
\mathbb{K}_n &= \frac{C_n E[(X_n - \hat{X}_{n|n-1})^2]}{K_n^2 + C_n^2 E[(X_n - \hat{X}_{n|n-1})^2]} \\
&= \frac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}}, \quad \text{where } V_{n|n-1} = E[(X_n - \hat{X}_{n|n-1})^2]. \quad \square
\end{aligned}
$$

This result is consistent with the Kalman gain derived in equation (3.23).

Chapter 4

The Use of Kalman Filter

4.1 Innovation Representation of Kalman Filter

Now, we modify the state equation by introducing a deterministic control term, thereby connecting the pure state estimation model to a control system model.

$$X_{n+1} = A_n X_n + B_n u_n + H_n \epsilon_n$$

$$Y_n = C_n X_n + K_n \eta_n \quad \forall n \geq 1$$

Where A_n, B_n, H_n, C_n, K_n are deterministic real values, and $(\epsilon_n)_{n\geq 0}$, $(\eta_n)_{n\geq 1}$ are iid sequences of $N(0,1)$. $X_0 \sim N(0, \sigma^2), Y_0 = 0$ and $X_0, Y_0, \epsilon_0, \epsilon_1, \ldots, \eta_l, \eta_2, \ldots$ are independent. u_n is a deterministic control input at time n, which is applied to system in time $[n, n+1)$.

All other assumptions remain unchanged, allowing us to proceed with the prediction and correction steps as described in Section 3.4.2, which are omitted here for brevity. The key distinction is in the prediction estimator shown in equation (4.1), expressed as $\hat{X}_{n|n-1} = A_{n-1}\hat{X}_{n-1|n-1} + B_{n-1}u_{n-1}$.

Below, we present the key equations corresponding to this modified state-space model.

1. **Prediction step**

$$\hat{X}_{n|n-1} = A_{n-1}\hat{X}_{n-1|n-1} + B_{n-1}u_{n-1} \tag{4.1}$$

$$V_{n|n-1} = A_{n-1}^2 V_{n-1|n-1} + H_{n-1}^2 \tag{4.2}$$

2. **Correction step**

$$\hat{X}_{n|n} = \hat{X}_{n|n-1} + \mathbb{K}_n(Y_n - C_n\hat{X}_{n|n-1}) \tag{4.3}$$

$$V_{n|n} = (1 - \mathbb{K}_n C_n)V_{n|n-1} \tag{4.4}$$

$$\mathbb{K}_n = \frac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}} \tag{4.5}$$

In the context of estimation, the prediction error defined in Section 3.4.4.1 ($\tilde{Y}_{n|n-1} := Y_n - \hat{Y}_{n|n-1}$), is the difference between the actual observation and the predicted observation. This error represents the new information derived from the latest observation that was not predicted by the filter, hence it is referred to as the **innovation** of the filter. It is often denoted by

$$\nu_n := \tilde{Y}_{n|n-1} = Y_n - \hat{Y}_{n|n-1} = Y_n - C_n\hat{X}_{n|n-1}$$

The innovation plays a crucial role: on one hand, it helps update the prediction estimator to a correction estimator, as shown in equation (4.3). On the other hand, it can serve as white noise, which is important in Linear Quadratic Gaussian (LQG) problems, a topic that will be introduced in a later section.

Property 4.1.1 (Innovation)**.** *The innovation process $(\nu_n)_{n\geq 0}$ is a Gaussian white noise process. It has a zero mean, $E[\nu_n] = 0$, and a variance given by $E[\nu_n^2] = C_n^2 V_{n|n-1} + K_n^2$ for $n \geq 0$. Additionally, the covariance between different time points is zero, $E[\nu_n \nu_m] = 0$ for $n \neq m$, ensuring that the innovations are uncorrelated (i.e. it's white).*

Proof. The mean of the innovation is

$$\begin{aligned} E[\nu_n] &= E[E[\nu_n \mid \mathcal{F}_{n-1}]] = E[E[Y_n - \hat{Y}_{n|n-1} \mid \mathcal{F}_{n-1}]] \\ &= E[\hat{Y}_{n|n-1} - \hat{Y}_{n|n-1}] = 0. \end{aligned}$$

The variance of the innovation is computed in the proof to get equation (3.25).

$$E[\nu_n^2] = C_n^2 V_{n|n-1} + K_n^2.$$

Here are some equations use to compute the covariance.

1.

$$\begin{aligned} \nu_n &= Y_n - \hat{Y}_{n|n-1} = C_n(X_n - \hat{X}_{n|n-1}) + K_n\eta_n \\ &= C_n\tilde{X}_n + K_n\eta_n \end{aligned} \tag{4.6}$$

2.

$$\begin{aligned} \tilde{X}_{n|n-1} &= X_n - \hat{X}_{n|n-1} \\ &= (A_{n-1}X_{n-1} + B_{n-1}u_{n-1} + H_{n-1}\epsilon_{n-1}) \\ &\quad - (A_{n-1}\hat{X}_{n-1|n-1} + B_{n-1}u_{n-1}) \\ &= A_{n-1}\tilde{X}_{n-1|n-1} + H_{n-1}\epsilon_{n-1} \end{aligned} \tag{4.7}$$

3.

$$\begin{aligned} \tilde{X}_{n|n} &= X_n - \hat{X}_{n|n} \\ &= X_k - \left(\hat{X}_{n|n-1} + \mathbb{K}_n\nu_n\right) \\ &= X_k - \left(\hat{X}_{n|n-1} + \mathbb{K}_n(C_n\tilde{X}_{n|n-1} + K_n\eta_n)\right), \text{ from equation (4.6)} \\ &= (1 - \mathbb{K}_nC_n)\tilde{X}_{n|n-1} - \mathbb{K}_n\eta_n \end{aligned} \tag{4.8}$$

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To consider $m = n + 1$ and omit the general case.

$$
\begin{aligned}
E[\nu_n \nu_{n+1}] &= E\left[\left(C_n \tilde{X}_{n|n-1} + K_n \eta_n\right)\left(C_{n+1}\tilde{X}_{n+1|n} + K_{n+1}\eta_{n+1}\right)\right] \\
&= \underbrace{E\left[C_n \tilde{X}_{n|n-1} \cdot C_{n+1}\tilde{X}_{n+1|n}\right]}_{①} + \underbrace{E\left[C_n \tilde{X}_{n|n-1} \cdot K_{n+1}\eta_{n+1}\right]}_{②} + \underbrace{E\left[C_{n+1}\tilde{X}_{n+1|n} \cdot K_n \eta_n\right]}_{③} + \\
&\underbrace{E\left[K_n \eta_n \cdot K_{n+1}\eta_{n+1}\right]}_{④}.
\end{aligned}
$$

Let's compute each term respectively:

-

$$
\begin{aligned}
① &= C_n C_{n+1} E[\tilde{X}_{n|n-1}\tilde{X}_{n+1|n}] \\
&= C_n C_{n+1} E[\tilde{X}_{n|n-1}(A_n \tilde{X}_{n|n} + H_n \epsilon_n)], \text{ from equation (4.7)} \\
&= C_n C_{n+1}\left(E[\tilde{X}_{n|n-1}A_n\tilde{X}_{n|n}] + E[\tilde{X}_{n|n-1}H_n\epsilon_n]\right) \\
&= C_n C_{n+1} E[\tilde{X}_{n|n-1}A_n E[(1 - \mathbb{K}_n C_n)\tilde{X}_{n|n-1} - \mathbb{K}_n \eta_n | \mathcal{F}_{n-1}]] \quad \text{, from equation(4.8),} \\
&\quad \text{, since } H_n\epsilon_n \text{ is independent of } \tilde{X}_{n|n-1} \text{ the second term is } 0. \\
&= C_n C_{n+1} E\left[\tilde{X}_{n|n-1}A_n\left(E[(1 - \mathbb{K}_n C_n)\tilde{X}_{n|n-1}] - E[\mathbb{K}_n \eta_n]\right)\right] \\
&\quad \text{, since } \tilde{X}_{n|n-1} \text{ and } \eta_n \text{ are independent of } \mathcal{F}_{n-1}. \\
&= C_n C_{n+1} E\left[\tilde{X}_{n|n-1}A_n \cdot 0\right] \quad \text{, since } \tilde{X}_{n|n-1} \text{ and } \eta_n \text{ have zero mean.} \\
&= 0
\end{aligned}
$$

- ② $= EE[C_n \tilde{X}_{n|n-1} \cdot K_{n+1}\eta_{n+1} \mid \mathcal{F}_{n-1}] = E[C_n \tilde{X}_{n|n-1} \cdot K_{n+1}E[\eta_{n+1} \mid \mathcal{F}_{n-1}]] = E[C_n \tilde{X}_{n|n-1} \cdot K_{n+1}E[\eta_{n+1}]] = 0$, by property 3.4.3, ν_{n+1} is independent of $\mathcal{F}_n$, and also $E[\eta_{n+1}] = 0$.

- ③$= E[C_{n+1}\tilde{X}_{n+1|n}] \cdot E[K_n \eta_n]$,since η_n is independent of $\tilde{X}_{n+1|n}$.

- ④ $= K_n K_{n+1} E[\eta_n \eta_{n+1}] = 0$, since $(\eta_n)_{n \geq 0}$ is independent.

And innovation is Gaussian because it is a linear combination of independent Gaussian random variables.

□

Theorem 4.1.1 (Kalman Theorem - Innovation Representation)**.** *There is an alternative representation of the Kalman filter, expressed in terms of prediction estimators only.*

$\hat{X}_{n|n-1}$ *satisfies the recursive equation:*

$$\begin{aligned} \hat{X}_{n+1|n} &= A_n \hat{X}_{n|n-1} + B_n u_n + \mathbb{K}_n^* \nu_n \quad , n \geq 0 \\ \hat{X}_{0|-1} &= 0, \end{aligned} \tag{4.9}$$

where

$$\mathbb{K}_n^* = A_n \mathbb{K}_n = \frac{A_n C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}}. \tag{4.10}$$

$V_{n|n-1}$ *satisfies the Riccati recursion for scalar values:*

$$\begin{aligned} V_{n+1|n} &= A_n^2 V_{n|n-1} + H_n^2 - \frac{A_n^2 C_n^2 V_{n|n-1}^2}{K_n^2 + C_n^2 V_{n|n-1}} \quad , n \geq 0 \\ V_{0|-1} &= \sigma^2. \end{aligned} \tag{4.11}$$

Proof. This can be easily derived from the key equations in the Prediction step and Correction step.

From the prediction estimator, the equation (4.1) :

$$\begin{aligned} \hat{X}_{n+1|n} &= A_n \hat{X}_{n|n} + B_n u_n \\ &= A_n \left(\hat{X}_{n|n-1} + \mathbb{K}_n \nu_n \right) + B_n u_n \quad \text{, replace } \hat{X}_{n|n} \text{ by relation (4.3).} \\ &= A_n \hat{X}_{n|n-1} + \mathbb{K}_n^* \nu_n + B_n u_n \quad \text{, where } \mathbb{K}_n^* := A_n \mathbb{K}_n. \end{aligned}$$

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Similarly, start from the prediction variance , equation (4.2) :

$$\begin{aligned} V_{n+1|n} &= A_n^2 V_{n|n} + H_n^2 \\ &= A_n^2 \left((1 - \mathbb{K}_n C_n) V_{n|n-1}\right) + H_n^2 \quad \text{, replace } V_{n|n} \text{ by relation (4.4).} \\ &= A_n^2 \left((1 - \frac{C_n V_{n|n-1}}{K_n^2 + C_n^2 V_{n|n-1}} C_n) V_{n|n-1} \right) + H_n^2 \\ &= A_n^2 + H_n^2 - \frac{A_n^2 C_n^2 V_{n|n-1}^2}{K_n^2 + C_n^2 V_{n|n-1}}. \end{aligned}$$

□

Remark 7. *In optimal control, the control policy often takes a form similar to the prediction estimator update in the Kalman filter. By defining* $\mathbb{K}_n^*$ *as* $A_n \mathbb{K}_n$ *, we make the structure of the Kalman filter's update step more analogous to the control gain in a control policy, see in Section 4.2.2..*

4.2 Control problem

Another class of problems triggered from state-space models is the control problem. It aims to determine the optimal input to drive the system to a desired state or output. Here we focus on minimizing a quadratic cost function. Starting with the deterministic case, this is known as the Linear Quadratic Regulator (LQR) problem. In the real world, it is more appropriate to consider noise, leading to the Stochastic Linear Quadratic Regulator (SLQR) problem, assuming complete state information (i.e., system states are available). Finally, when the state cannot be observed directly (incomplete state information), the problem becomes the Linear Quadratic Gaussian (LQG) problem, where the Kalman filter plays an important role in providing linear quadratic estimation.

4.2.1 Linear Quadratic Regulator (LQR)

Consider the linear state-space model with a control input. In this context, we assume that the system states are fully observable and available at each time step, thereby eliminating the

need for an observation equation. This assumption simplifies the problem by focusing solely on the control aspect without the complications of state estimation.

Let the linear system be described by state-space model (S).
For $n \geq 1$,

$$S: \qquad x_{n+1} = A_n x_n + B_n u_n \tag{4.12}$$
$$\text{with initial condition } x_0.$$

Where x_n is the system state at time n, and u_n is the control input applied to the system during the time interval $[n, n+1)$. A_n, B_n are deterministic values in $\mathbb{R}$.

Let the **cost function** (or performance function) (J) over the time interval $[0, N]$ be defined as

$$J: \qquad J = \sum_{i=0}^{N-1} (Q_i x_i^2 + R_i u_i^2) + Q_f x_N^2.$$

Where $Q, Q_f \geq 0$ and $R > 0$. This function quantifies the performance of the control strategy. The number of stages,N, is referred to as **the time horizon**. We specifically address the finite-horizon problem where N is finite.

In the Linear Quadratic Regulator (LQR) problem, the term "Linear" refers to the system dynamics (S) satisfying a linear function, while "Quadratic" refers to the cost function (J) being in quadratic form. The objective is to find a sequence of control inputs $(u_0, \ldots, u_{N-1})$ that minimizes the cost function. The way to help us to break down this complex optimization problem into simpler subproblems is the Principle of Optimality.

For an intermediate stage k, define **the cost-to-go** from state x at stage k as

$$J_k: \qquad J_k = \sum_{i=k}^{N-1} \left(Q_i x_i^2 + R_i u_i^2\right) + Q_f x_N^2.$$

If we can solve these subproblems optimally, then we can construct an overall optimal solution for (S, J).

This series of following theorems below (Theorems 4.2.1, 4.2.2, and 4.2.3) is presented to provide a more comprehensive understanding of the topic, but the detailed proofs are omitted in this thesis to maintain focus on the broader discussion. Readers can consult Caines's *Linear Stochastic Systems* (1988, pp. 661-665) [8]. The proof included here is those directly connected to the derivation of the solution of the LQR problem.

Theorem 4.2.1 (The Principle of Optimality)**.** *Assume that there exists an optimal control sequence* $(u_k^*, u_{k+1}^*, \ldots, u_N^*)$ *exists for* (S, J_k) *with the initial condition* x *at time* k *(i.e.* $x_k = x$*). For a later time* $k' \geq k$*, the sequence* $(u_{k'}^*, u_{k'+1}^*, \ldots, u_N^*)$ *is optimal for* $(S, J_{k'})$ *with the initial condition* $x_{k'}^*$ *at time* k'*, where* $x_{k'}^* = x_{k'}(x, k, u^*)$*.*

Proof. The proof is omitted. □

For the minimal value corresponding to (S, J_k) with the initial condition x at time k (i.e. $x_k = x$), we define the value function at each time k as

$$\begin{aligned} V_k(x) &:= \min_{u_k, \ldots, u_{N-1}} J_k \\ &= \min_{u_k, \ldots, u_{N-1}} \left\{ \sum_{i=k}^{N-1} (Q_i x_i^2 + R_i u_i^2) + Q_f x_N^2 \right\} \end{aligned}$$

subject to the system dynamics $S : x_{i+1} = A_i x_i + B_i u_i$ for $i = k, \ldots, N-1$.

This value function represents the minimum cost-to-go from time k to the final time N, starting from state $x_k = x$.

Theorem 4.2.2 (The Dynamic Programming Equation)**.** *Assume that an optimal control law exists for* (S, J_k) *on the interval* $[0, N]$*. Then the value function* $V_k(x_k)$ *satisfies the Dynamic Programming Equation:*

$$V_k(x_k) = \min_{u_k} \left\{ Q_k x_k^2 + R_{u_k}^2 + V_{k+1}(x_{k+1}) \right\} \quad \textit{where } x_k \in \mathbb{R},\ k = N-1, \ldots, 0. \tag{4.13}$$

$$V_N(x_N) = Q_f x_N^2 \quad x_N \in \mathbb{R},\ \textit{is the terminal condition.} \tag{4.14}$$

Equation (4.13) is known as the Bellman equation. It defines the value function $V_k(x_k)$ through recursive minimization over control inputs, guiding the derivation of the optimal control law. In continuous-time problems, it's referred to as the Hamilton-Jacobi-Bellman (HJB) equation, and it's central to solve complex control problems by breaking them into manageable steps.

Proof. Start from the definition of the value function, and it's done by separate the first stage of the summation (corresponding to time k) from the remaining terms.

$$\begin{aligned}
V_k(x_k) &:= \min_{u_k,\dots,u_{N-1}} \Big\{ \sum_{i=k}^{N-1} (Q_i x_i^2 + R_i u_i^2) + Q_f x_N^2 \Big\} \\
&= \min_{u_k} \Big\{ \min_{u_{k+1},\dots,u_{N-1}} \Big\{ Q_k x_k^2 + R_k u_k^2 + \sum_{i=k+1}^{N-1} (Q_i x_i^2 + R_i u_i^2) + Q_f x_N^2 \Big\} \Big\} \\
&= \min_{u_k} \Big\{ Q_k x_k^2 + R_k u_k^2 + \min_{u_{k+1},\dots,u_{N-1}} \Big\{ \sum_{i=k+1}^{N-1} (Q_i x_i^2 + R_i u_i^2) + Q_f x_N^2 \Big\} \Big\} \\
&= \min_{u_k} \Big\{ Q_k x_k^2 + R_k u_k^2 + V_{k+1}(x_{k+1}) \, ; x_{k+1} = A_k x_k + B_k u_k \Big\}
\end{aligned}$$

□

Theorem 4.2.3 (The Verification Theorem)**.** *Let u_k° and V_k° satisfy the Dynamic Programming Equation. That is*

$$\begin{aligned}
V_k^\circ(x_k) &= \min_{u_k} \{ Q_k x_k^2 + R_k u_k^2 + V_{k+1}^\circ(x_{k+1}) \} \\
&= Q_k x_k^2 + R_k (u_k^\circ)^2 + V_{k+1}^\circ \left(x_{k+1}(x_k, u_k^\circ) \right) \quad \textit{where } x_k \in \mathbb{R},\ k = N-1, \dots, 0, \\
&\textit{and } x_{k+1}(x_k, u_k^\circ) = A_k x_k + B_k u_k^\circ.
\end{aligned}$$

$V_N^\circ(x_N) = Q_f x_N^2 \quad x_N \in \mathbb{R}$, *is the terminal condition.*

Then the sequence $(u_0^\circ, \dots, u_N^\circ)$ is optimal for the system (S, J_k) on $[0, N]$ and the optimal value function V_k exists and $V_k = V_k^\circ$.

Proof. The proof is omitted. □

The Verification Theorem is a key result in optimal control theory, as it provides a crucial

method to confirm that a candidate control policy is indeed optimal and achieves the minimal cost. This theorem is particularly valuable when directly determining the optimal policy is challenging. By ensuring that the conditions for optimality are satisfied, the Verification Theorem guarantees that the derived policy is the best possible solution without the need for exhaustive exploration of all alternatives.

Theorem 4.2.4 (LQR solution)**.** *An optimal control sequence exists for system* (S, J_k) *over a finite time horizon* $[0, N]$.

- *The solution to the Bellman equation (4.13) and (4.14) for the linear regular problem* (S, J_k) *is given by*

$$V_k(x_k) = S_k x_k^2 \quad \text{for } k = N, \dots, 0.$$

where $S_k,\ k = N-1, \dots, 0$ *satisfies the scalar version of Riccati recursion*

$$S_k = Q_k + A_k^2 S_{k+1} - \frac{A_k^2 B_k^2 S_{k+1}^2}{R_k + B_k^2 S_{k+1}} \quad \text{and} \quad S_N := Q_f. \tag{4.15}$$

- *The optimal control input is given in the state feedback form*

$$u_k^* = -L_k x_k\ , \tag{4.16}$$

where the control gain L_k *is defined as*

$$L_k := \frac{A_k B_k S_{k+1}}{R_k + B_k^2 Q_f}. \tag{4.17}$$

- *The ultimate optimal cost is* $J^* = V_0(x_0) = S_0 x_0^2$.

Proof. Start from $V_N(x_N) = Q_f x_N^2$.
Then do the recursion step (from N to $N-1$)

$$\begin{aligned} V_{N-1}(x_{N-1}) &= \min_{u_{N-1}} \left\{ Q_{N-1} x_{N-1}^2 + R_{N-1} u_{N-1}^2 + Q_f x_N^2 \right\} \\ &= \min_{u_{N-1}} \left\{ Q_{N-1} x_{N-1}^2 + R_{N-1} u_{N-1}^2 + Q_f (A x_{N-1} + B u_{N-1})^2 \right\} \end{aligned}$$

, replace x_N using system dynamic (4.12).

To find the optimal control u_{N-1}^* , take the derivative of the cost function with respect to u_{N-1}. Set

$$\frac{\partial}{\partial u_{N-1}}\left\{Q_{N-1}x_{N-1}^2 + R_{N-1}u_{N-1}^2 + Q_f(A_{N-1}x_{N-1} + B_{N-1}u_{N-1})^2\right\} = 0.$$

Then expanding and differentiating, get

$$2R_{N-1}u_{N-1} + 2Q_fB_{N-1}(A_{N-1}x_{N-1} + B_{N-1}u_{N-1}) = 0.$$

Hence the optimal control is

$$\begin{aligned} u_{N-1}^* &= -\frac{A_{N-1}B_{N-1}Q_f}{R_{N-1} + B_{N-1}^2Q_f}\, x_{N-1} \\ &= L_{N-1}\, x_{N-1} \quad \text{where } L_{N-1} \text{ is the control gain.} \end{aligned}$$

Substituting the optimal control $u_{N-1}^* = -L_{N-1}x_{N-1}$ into the value function to obtain $V_{N-1}(x_{N-1})$,

$$\begin{aligned} V_{N-1}(x_{N-1}) &= Q_{N-1}x_{N-1}^2 + R_{N-1}\left(\frac{A_{N-1}B_{N-1}Q_f}{R_{N-1} + B_{N-1}^2Q_f}x_{N-1}\right)^2 \\ &\quad + Q_f\left(A_{N-1}x_{N-1} - \frac{B_{N-1}^2A_{N-1}Q_f}{R_{N-1} + B_{N-1}^2Q_f}x_{N-1}\right)^2 \\ &= x_{N-1}^2\left(Q_{N-1} + \underbrace{R_{N-1}\left(\frac{A_{N-1}B_{N-1}Q_f}{R_{N-1} + B_{N-1}^2Q_f}\right)^2}_{①} + \underbrace{Q_f\left(A_{N-1} - \frac{B_{N-1}^2A_{N-1}Q_f}{R_{N-1} + B_{N-1}^2Q_f}\right)^2}_{②}\right) \\ &= x_{N-1}^2\left(Q_{N-1} + \frac{R_{N-1}A_{N-1}^2B_{N-1}^2Q_f^2}{(R_{N-1} + B_{N-1}^2Q_f)^2} + \frac{A_{N-1}^2Q_fR_{N-1}^2}{(R_{N-1} + B_{N-1}^2Q_f)^2}\right) \\ &= x_{N-1}^2\left(Q_{N-1} + \frac{A_{N-1}^2(R_{N-1}B_{N-1}^2Q_f^2 + Q_fR^2)}{(R_{N-1} + B_{N-1}^2Q_f)^2}\right) \\ &= x_{N-1}^2\left(Q_{N-1} + \frac{A_{N-1}^2R_{N-1}Q_f(B_{N-1}^2Q_f + R_{N-1})}{(R_{N-1} + B_{N-1}^2Q_f)^2}\right) \\ &= x_{N-1}^2\left(Q_{N-1} + A_{N-1}^2Q_f - \frac{A_{N-1}^2B_{N-1}^2Q_f^2}{R_{N-1} + B_{N-1}^2Q_f}\right) \quad \square \end{aligned}$$

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Where

$$
\begin{aligned}
① &= \frac{R_{N-1}A_{N-1}^2B_{N-1}^2Q_f^2}{(R_{N-1}+B_{N-1}^2Q_f)^2}.\\
② &= Q_f\left(A_{N-1}-\frac{B_{N-1}^2A_{N-1}Q_f}{R_{N-1}+B_{N-1}^2Q_f}\right)^2 = Q_f\left(\frac{A_{N-1}(R_{N-1}+B_{N-1}^2Q_f)-B_{N-1}^2A_{N-1}Q_f}{R_{N-1}+B_{N-1}^2Q_f}\right)^2.\\
&= Q_f\left(\frac{A_{N-1}R_{N-1}}{R_{N-1}+B_{N-1}^2Q_f}\right)^2 = \frac{A_{N-1}^2Q_fR^2}{(R_{N-1}+B_{N-1}^2Q_f)^2}.
\end{aligned}
$$

The final expression for $V_{N-1}(x_{N-1})$ confirms the quadratic form $S_{N-1}x_{N-1}^2$, where S_{N-1} is derived as:

$$S_{N-1} := Q_{N-1} + A_{N-1}^2Q_f - \frac{A_{N-1}^2B_{N-1}^2Q_f^2}{R_{N-1}+B_{N-1}^2Q_f}.$$

Then applying the same induction process for the transition from arbitrary $k+1$ to k, $k = N-1,\ldots,0$, thereby arriving at the final conclusion.

□

4.2.2 Duality

The duality between the Kalman filter and control theory arises from the mathematical similarities between the optimal estimation problem and the optimal control problem. The Riccati equation in the LQR problem is mathematically similar to the Riccati equation in the Kalman filter. Also the control gain in the optimal control problem is analogous to the Kalman gain in the estimation problem.

The difference is that the Riccati recursion runs backwards from a terminal condition at time N, whereas the filtering Riccati recursion for the prediction variance $V_{n|n-1}$ runs forward from an initial condition at $n = 0$.

By appropriately relabeling and shifting the time indices, one can derive the optimal control solutions from the filtering problem and vice versa.

Recall from Theorem 4.2.4, **Optimal control problem:**

- The Riccati recursion: from equation (4.15)

$$S_n = Q_n + A_n^2 S_{n+1} - \frac{A_n^2 B_n^2 S_{n+1}^2}{R_n + B_n^2 S_{n+1}}, \quad n = N-1, \ldots, 0.$$
$$S_N = Q_f$$

- The control gain: from (4.17)
$$L_n = \frac{A_n B_n S_{n+1}}{R_n + B_n^2 S_{n+1}}$$

Recall from Theorem 4.1.1, **Kalman filter:**

For $n \geq 0$. Write $V_{n|n-1}$ simply as V_n and $V_{0|-1} = V_0$.

- The Riccati recursion: from equation (4.11)
$$V_{n+1} = H_n^2 + A_n^2 V_n - \frac{A_n^2 C_n^2 V_n^2}{K_n^2 + C_n^2 V_n}, \quad n \geq 0.$$
$$V_0 = \sigma^2$$

- The Kalman gain: from (4.10)
$$\mathbb{K}_n^* = \frac{A_n C_n V_n}{K_n^2 + C_n^2 V_n}$$

Filtering	**Control**
(time) n	$N-n$
A_n	A_{N-n}
C_n	B_{N-n}
H_n^2	Q_{N-n}
K_n^2	R_{N-n}
V_n	S_{N-n}
$\mathbb{K}_n^*$	L_{N-n}

Table 4.1: Correspondence between Filtering and Control Variables

If we take the filtering Riccati equation, relabel the variables according to the table, and shift the time index appropriately ($n \to N-n$), we obtain the control Riccati recursion.

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4.2.3 Stochastic Linear Quadratic Regulator (SLQR)

In many real-world applications, systems are influenced by random noise. So after discussing the deterministic LQR, it's important to consider systems affected by these random disturbances. This brings us to the Stochastic Linear Quadratic Regulator (SLQR) problem, where the state equation is modified to include noise. We first consider situations where the state equation includes noise and the state can be observed directly, meaning we have **complete state information**.

The state space model (S) is given by

$$S: \qquad X_{n+1} = A_n X_n + B_n u_n + H_n \epsilon_n \quad \text{,for } n \geq 1. \tag{4.18}$$

Where A_n, B_n, H_n are deterministic real values, and $(\epsilon_n)_{n\geq 0}$ are iid sequences of $N(0,1)$. $X_0 \sim N(0, \sigma^2)$ and X_0 is independent of $(\epsilon_n)_{n\geq 0}$.

At time n, the states $X_0, \ldots, X_n$ are available, but $H_n\epsilon_n$ is not. Due to the Markov nature of the system, the control input u_n is determined solely by the current state X_n, expressed as

$$u_n = \phi_n(X_n) \quad \text{, with } \phi_n : \mathbf{R} \mapsto \mathbf{R} \text{ as control policy.}$$

This approach is known as state feedback control.

The cost function (J) needs to account for the inherent randomness in the system. It is defined by

$$J: \qquad J = E\left[\sum_{i=0}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2\right].$$

over finite horizon $[0, N]$. Here, the expectation is taken over the initial state X_0 and the system noise (ϵ_n). The terms $Q, Q_f \geq 0$ and $R > 0$ ensure that the cost function penalizes large deviations in the state and excessive control effort.

The objective is to choose control policies $\phi_0, \ldots, \phi_{N-1}$ to minimize the cost function. As in the deterministic case, this can be achieved by considering simpler subproblems at each

intermediate stage.

Let the value function at time k, for $k = 0, \ldots, N$ with the initial condition $X_k = x$ be

$$V_k(x) := \min_{\phi_k,\ldots,\phi_{N-1}} \left\{ E_{k,x} \left[\sum_{i=k}^{N-1} (Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \right] \right\} \tag{4.19}$$
$$= \min_{\phi_k,\ldots,\phi_{N-1}} \left\{ E \left[\sum_{i=k}^{N-1} (Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \,\middle|\, X_k = x \right] \right\},$$

where $E_{k,x}$ denotes the expectation given that the process begins at $X_k = x$, and the expectation is taken over the noise $\{\epsilon_k, \ldots, \epsilon_{N-1}\}$.
Subject to $S : X_{i+1} = A_i X_i + B_i u_i + H_i \epsilon_i$ and $u_i = \phi_i(X_i)$ for $i = k, \ldots, N-1$.

There are corresponding theorems in the stochastic version analogous to Theorems 4.2.1, 4.2.2, and 4.2.3. In this context, only the verification that the value function $V_k(x)$ defined in equation (4.19) satisfies the Dynamic Programming Equation is presented. Details can be found in Caines's *Linear Stochastic Systems* (1988, pp. 672-676) [8].

$$V_k(x_k) = \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E_{k,x_k} \left[V_{k+1}(x_{k+1}) \right] \right\} \quad \text{where } x_k \in \mathbb{R},\ k = N-1, \ldots, 0. \tag{4.20}$$

$$V_N(x_N) = Q_f x_N^2 \quad \text{, where } x_N \in \mathbb{R}, \text{ is the terminal condition.} \tag{4.21}$$

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Proof.

$$
\begin{aligned}
V_k(x_k) &:= \min_{\phi_k,\ldots,\phi_{N-1}} \left\{ E_{k,x_k}\left[\sum_{i=k}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2\right]\right\} \\
&= \min_{\phi_k,\ldots,\phi_{N-1}} \left\{ E\left[\sum_{i=k}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k,\ldots,\phi_{N-1}} \left\{ E\left[Q_k X_k^2 + R_k u_k^2 + E\left[\sum_{i=k+1}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \,\middle|\, X_{k+1} = x_{k+1}\right] \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k} \left\{ E\left[Q_k X_k^2 + R_k u_k^2 + \min_{\phi_{k+1},\ldots,\phi_{N-1}} \left\{ E\left[\sum_{i=k+1}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \,\middle|\, X_{k+1} = x_{k+1}\right]\right\} \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E\left[\min_{\phi_{k+1},\ldots,\phi_{N-1}} \left\{ E\left[\sum_{i=k+1}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2 \,\middle|\, X_{k+1} = x_{k+1}\right]\right\} \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E\left[\min_{\phi_{k+1},\ldots,\phi_{N-1}} \left\{ E_{k+1,x_{k+1}}\left[\sum_{i=k+1}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2\right]\right\} \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E\left[V_{k+1}(x_{k+1}) \,\middle|\, X_k = x_k\right]\right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E_{k,x_k}\left[V_{k+1}(x_{k+1})\right]\right\} \quad \square
\end{aligned}
$$

□

Theorem 4.2.5 (SLQR solution)**.** *For the stochastic linear regulator with complete state information, an optimal control sequence exists for system* (S, J_k) *over a finite time horizon* $[0, N]$.

- *The solution to the Bellman equations (4.20) and (4.21) is given by*

$$V_k(X_k) = S_k X_k^2 + q_k \quad \textit{for } k = N, \ldots, 0.$$

where $S_k,\ k = N-1, \ldots, 0$ *satisfies the discrete-time Riccati recursion*

$$S_k = Q_k + A_k^2 S_{k+1} - \frac{A_k^2 B_k^2 S_{k+1}^2}{R_k + B_k^2 S_{k+1}} \quad \textit{and} \quad S_N := Q_f. \tag{4.22}$$

$$q_k = S_{k+1} H_k^2 + q_{k+1} \quad \textit{and} \quad q_N := 0. \tag{4.23}$$

- *The optimal control policy is identical to the deterministic case.*

$$u_k^* = \phi_k^*(X_k) = -L_k X_k$$

where

$$L_k = \frac{A_k B_k S_{k+1}}{R_k + B_k^2 Q_f} \quad \textit{is given by equation (4.17).}$$

- *The ultimate optimal cost is*

$$J^* = E\left[V_0(X_0)\right] = S_0 E\left[X_0^2\right] + \sum_{i=0}^{N-1} S_{i+1} H_i^2. \tag{4.24}$$

Proof. The proof follows a similar approach to the deterministic case presented in Theorem 4.2.4.

The terminal cost is $V_N(x_N) = Q_f x_N^2$ with $q_N = 0$. The step from N to $N-1$ has already been addressed in Theorem 4.2.4. Here, we proceed with the step from $k+1$ to k for $k = N-1, \ldots, 0$.

Assume $V_{k+1}(X_{k+1}) = S_{k+1} X_{k+1}^2 + q_{k+1}$ is valid.

$$\begin{aligned}
V_k(x_k) &= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E_{k,x_k}\left[V_{k+1}(x_{k+1})\right] \right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E_{k,x_k}\left[S_{k+1} X_{k+1}^2 + q_{k+1}\right] \right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + E_{k,x_k}\left[S_{k+1}(A_k X_k + B_k u_k + H_k \epsilon_k)^2 + q_{k+1}\right] \right\} \\
&\qquad \text{, replace } X_{k+1} \text{ by using dynamic (4.18)} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + S_{k+1} E_{k,x_k} \left[\underbrace{(A_k X_k + B_k u_k)^2}_{①} + \underbrace{2(A_k X_k + B_k u_k)(H_k^2 \epsilon_k^2)}_{②} + \underbrace{H_k^2 \epsilon_k^2}_{③} \right] + q_{k+1} \right\} \\
&= \min_{\phi_k} \left\{ Q_k X_k^2 + R_k u_k^2 + S_{k+1}(A_k x_k + B_k u_k)^2 + \underbrace{S_{k+1} H_k^2 + q_{k+1}}_{=:\, q_k} \right\}
\end{aligned}$$

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Where

$$① = E[(A_kX_k + B_ku_k)^2 \mid X_k = x_k] = (A_kx_k + B_ku_k)^2 \quad \text{, since } u_k = \phi_k(X_k).$$

$$② = E[2(A_kX_k + B_ku_k)(H_k^2\epsilon_k^2) \mid X_k = x_k] = 2(A_kx_k + B_ku_k)H_k^2E[\epsilon_k|X_k = x_k]$$

$$= 2(A_kx_k + B_ku_k)H_k^2E[\epsilon_k] = 0 \quad \text{since } \epsilon_k \text{ is independent of } X_k \text{ from property 3.4.2 and } \epsilon_k \sim N(0,1).$$

$$③ = H_k^2E[\epsilon_k^2|X_k = x_k] = H_k^2E[\epsilon_k^2] = H_k^2.$$

Define $q_k := S_{k+1}H_k^2 + q_{k+1}$ and it's a constant allowing us to take it out of the expectation. Then

$$V_k(x_k) = \min_{\phi_k} \left\{ Q_kX_k^2 + R_ku_k^2 + S_{k+1}(A_kx_k + B_ku_k)^2 \right\} + q_k.$$

Note that the function $V_k(x_k)$ follows the same recursive form as in the deterministic LQR, but with an additional constant term (q_k), yielding the same result.

Hence it follows that the optimal policy is the same as in equations (4.16) and (4.17),

$$u_k^* = \phi_k^*(X_k) = -L_kX_k \quad \text{,where } L_k = \frac{A_kB_kS_{k+1}}{R_k + B_k^2Q_f}.$$

Substituting the optimal control $u_{N-1}^* = -L_{N-1}x_{N-1}$ gives

$$V_k(X_k) = S_kX_k + q_k$$

with

$$S_k = Q_k + A_k^2S_{k+1} - \frac{A_k^2B_k^2S_{k+1}^2}{R_k + B_k^2S_{k+1}}$$

$$q_k = S_{k+1}H_k^2 + q_{k+1}.$$

The Riccati recursion for S_k is the same as equation (4.15).

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The ultimate optimal cost is computed as

$$\begin{aligned} J^* &= E\left[V_0(X_0)\right] \\ &= E\left[S_0 X_0^2 + q_0\right] \\ &= S_0 E\left[X_0^2\right] + \left(S_1 H_0^2 + q_1\right) \\ &= S_0 E\left[X_0^2\right] + \sum_{i=0}^{N-1} S_{i+1} H_i^2 \end{aligned}$$

where the expectation is taking over all X_0. □

4.2.4 Linear Quadratic Gaussian Regulator (LQG)

Now turning to a more complicated case, when the states X_n's cannot be observed directly, this situation is referred to as **incomplete state information**. So the state-space model must include an observation equation.

System dynamic is modeled by

$$S: \qquad X_{n+1} = AX_n + Bu_n + H\epsilon_n \tag{4.25}$$

$$Y_n = CX_n + K\eta_n, \quad \forall n \geq 1 \tag{4.26}$$

Where A, B, H, C, K are deterministic real values, and $(\epsilon_n)_{n\geq 0}$, $(\eta_n)_{n\geq 1}$ are iid sequences of $N(0,1)$. $X_0 \sim N(0, \sigma^2), Y_0 = 0$ and $X_0, Y_0, \epsilon_0, \epsilon_1, \ldots, \eta_l, \eta_2, \ldots$ are independent.

In the Linear Quadratic Gaussian (LQG) problem, the terms "Linear" and "Quadratic" carry the same meanings as in LQR problems. The "Gaussian" part indicates that the system noise and observation noise both follow a Gaussian distribution. This assumption of Gaussian noise enables the use of the Kalman filter.

Given this setup, while the true states X_n are not directly observable, we rely on the Kalman filter by providing the best state estimation through noisy observations. The separation principle states that the problems of state estimation and control can be separated and solved

independently. Specifically, in the LQG framework, this principle allows us to break the problem into two parts: state estimation, managed by the Kalman filter, and optimal control, where the estimated state is treated as if it were the true state, handled through the SLQR approach.

Here, the control input u_n depends on the observations $Y_1, \ldots, Y_{n-1}$, written as

$$u_n = \phi_n(Y_1, \ldots, Y_{n-1}) \quad \text{, with } \phi_n : \mathbf{R} \mapsto \mathbf{R} \text{ as control policy.}$$

This approach is known as output feedback control. The reason it does not include Y_n is that, in many real-time applications, control actions need to be applied immediately, and there isn't time to wait for the new observation at time n.

Using the Kalman filter, we treat the prediction estimator $\hat{X}_{n|n-1}$ as the true state. As before, we aim to choose control policies $\phi_0, \ldots, \phi_{N-1}$ to minimize the cost function (J), where

$$J: \qquad J = E\left[\sum_{i=0}^{N-1}(QX_i^2 + Ru_i^2) + Q_f X_N^2\right].$$

The randomness in the expectation is from X_0, $\{\epsilon_0, \ldots, \epsilon_{N-1}\}$, and $\{\eta_0, \ldots, \eta_{N-1}\}$, with $Q, Q_f \geq 0$ and $R > 0$.

Since the control input will be derived based on the prediction estimator $\hat{X}_{i|i-1}$, the cost function (J) must be reformulated in terms of $\hat{X}_{i|i-1}$ instead of the state X_i.

Lemma 4.2.1. *To minimize*

$$J = E\left[\sum_{i=0}^{N-1}(QX_i^2 + Ru_i^2) + Q_f X_N^2\right]$$

with respect to $\phi_0, \ldots, \phi_{N-1}$ is equivalent to minimizing

$$J^\circ: \qquad J^\circ = E\left[\sum_{i=0}^{N-1}(Q\hat{X}_{i|i-1}^2 + Ru_i^2) + Q_f \hat{X}_{N|N-1}^2\right]. \tag{4.27}$$

Proof. Start with $E[Q_i X_i^2]$.

$$\begin{aligned}
E[Q_i X_i^2] &= EE\left[Q_i X_i^2 \,\middle|\, Y_0, \ldots, Y_{i-1}\right] \\
&= EE\left[Q_i \left(\underbrace{X_i - \hat{X}_{i|i-1}}_{} + \hat{X}_{i|i-1}\right)^2 \,\middle|\, Y_0, \ldots, Y_{i-1}\right] \\
&= EE\left[Q_i \left(\underbrace{\tilde{X}_{i|i-1}}_{} + \hat{X}_{i|i-1}\right)^2 \,\middle|\, Y_0, \ldots, Y_{i-1}\right] \\
&= EQ_i E\left[\underbrace{\tilde{X}_{i|i-1}^2}_{①} + \underbrace{2\tilde{X}_{i|i-1}\hat{X}_{i|i-1}}_{②} + \underbrace{\hat{X}_{i|i-1}^2}_{③} \,\middle|\, Y_0, \ldots, Y_{i-1}\right] \\
&= E\left[Q_i V_{i|i-1} + \hat{X}_{i|i-1}^2\right] \\
&= Q_i V_{i|i-1} + E[Q_i \hat{X}_{i|i-1}^2]
\end{aligned}$$

$① = EE[\tilde{X}_{i|i-1}^2 \,|Y_0, \ldots, Y_{i-1}] = E[\tilde{X}_{i|i-1}^2] = V_{i|i-1}$

$② = 2EE[\tilde{X}_{i|i-1}\hat{X}_{i|i-1} \,|Y_0, \ldots, Y_{i-1}] = 2EE[\tilde{X}_{i|i-1}\hat{X}_{i|i-1}] = 0$ since they are independent.

$③ = EE[\hat{X}_{i|i-1}^2 \,|Y_0, \ldots, Y_{i-1}] = E[\hat{X}_{i|i-1}^2]$

Then

$$\begin{aligned}
J &:= E\left[\sum_{i=0}^{N-1}(Q_i X_i^2 + R_i u_i^2) + Q_f X_N^2\right] \\
&= E\left[\sum_{i=0}^{N-1}(Q_i \hat{X}_{i|i-1}^2 + R_i u_i^2) + Q_f \hat{X}_{N|N-1}^2\right] + \sum_{i=0}^{N-1} Q_i V_{i|i-1} + Q_f V_{N|N-1} \\
&= J^\circ + \sum_{i=0}^{N-1} Q_i V_{i|i-1} + Q_f V_{N|N-1} \qquad (4.28)
\end{aligned}$$

The term $\sum_{i=0}^{N-1} Q_i V_{i|i-1} + Q_f V_{N|N-1}$ is independent of the control inputs u_i's. Therefore, minimizing J is equivalent to minimizing J°. □

This equivalence transforms the problem into finding the optimal control policies $\phi_0, \ldots, \phi_{N-1}$ with respect to the modified system (S, J°). In this setting, the prediction estimator

serves as true state, which $\hat{X}_{i|i-1}$ is available, with dynamics (4.9) :

$$\hat{X}_{i+1|i} = A_i \hat{X}_{i|i-1} + B_i u_i + \mathbb{K}_i^* \nu_i \tag{4.29}$$

derived from the Innovation representation of Kalman filter (Theorem 4.1.1), where $\nu_i = Y_i - C\hat{X}_{i|i-1}$.

Moreover, by property 4.1.1 , the innovation process $(\nu_n)_{n\geq 0}$ is an independent Gaussian process with zero mean and variance $C_n^2 V_{n|n-1} + K_n^2$. So in equation (4.29), $\mathbb{K}_i^* \nu_i$ can serve as the system noise, allowing us to revert to the case where the system is complete state information. Then we can use the results derived before.

Theorem 4.2.6. *The optimal policy for (S, J) in LQG setting is*

$$u_k^* = \phi_k(Y_0, \ldots, Y_{k-1}) = -L_k \hat{X}_{k|k-1},$$

where L_k is given as equation (4.17) of the deterministic case and so as in SLQR case. The ultimate optimal cost is

$$J^* = (J^\circ)^* + \sum_{i=0}^{N-1} Q_i V_{i|i-1} + Q_f V_{N|N-1},$$

where

$$(J^\circ)^* = S_0 E\left[X_0^2\right] + \sum_{i=0}^{N-1} S_{i+1} (\mathbb{K}_{i+1}^*)^2 \left(C_i^2 V_{i|i-1} + K_i^2\right).$$

Proof. By Lemma 4.2.1, it's equivalent to find the optimal control policy with respect to (S, J°). Then, treat the prediction estimator of the state $\hat{X}_{i|i-1}$ as the true state with dynamic (4.29):

$$\hat{X}_{i+1|i} = A_i \hat{X}_{i|i-1} + B_i u_i + \mathbb{K}_i^* \nu_i$$

, where $\mathbb{K}_i^* \nu_i$ represents the system noise. Returning to the SLQR case, by Theorem 4.2.5, the optimal control policy is $u_k^* = -L_k \hat{X}_{k|k-1}$. Referring to the result (4.24) in Theorem 4.2.5, the variance of the noise is $(\mathbb{K}_{i+1}^*)^2 \left(C_i^2 V_{i|i-1} + K_i^2\right)$.

Thus, the ultimate optimal cost of J° is

$$(J^{\circ})^{*} = S_0 \mathbb{E}\left[X_0^2\right] + \sum_{i=0}^{N-1} S_{i+1} (\mathbb{K}_{i+1}^{*})^2 \left(C_i^2 V_{i|i-1} + K_i^2\right).$$

Also referring to equation (4.28), we obtain the desired ultimate optimal cost of J:

$$J^{*} = (J^{\circ})^{*} + \sum_{i=0}^{N-1} Q_i V_{i|i-1} + Q_f V_{N|N-1}.$$

□

This theorem states that the ultimate optimal cost J^{*} is divided into two parts. The first part, $(J^{\circ})^{*}$, represents the cost as if the prediction estimates $\hat{X}_{i|i-1}$ were the actual state in the SLQR problem. The second part accounts for the additional cost associated with estimating the state.

In this section demonstrates one of the practical use of the Kalman filter in control problem. Kalman filter provides state estimator $\hat{X}_{k|k-1}$ to simplify the control problem. By treating these estimators as the true state, the theorem allows for the application of optimal control techniques as if complete state information were available. The Kalman filter thus plays a crucial role in bridging the gap between estimation and control, enabling effective decision-making in systems with incomplete or noisy observations.

In summary, this section highlights the application of the Kalman filter within the Linear Quadratic Gaussian (LQG) control framework. By leveraging the Kalman filter's state estimation, the LQG problem effectively reduces to a Stochastic Linear Quadratic Regulator (SLQR) problem, where the control policy is designed as if the prediction estimates were the actual states. The optimal control policy derived under these conditions not only minimizes the cost function but also elegantly separates the total cost into the performance cost, as if the estimates were accurate, and an additional cost reflecting the uncertainty in the state estimation. This dual approach underlines the strength of the Kalman filter in handling incomplete state information, ensuring that control decisions are both practical and robust in the face of uncertainty.

Chapter 5

Conclusion

In this thesis, the exploration of martingales offers a deeper understanding of the convergence properties of random processes. This insight facilitates a clearer grasp of the foundational concepts in recursive estimation, which naturally leads to the study of the Kalman-Bucy filter, one of the most fundamental cases in filtering theory. The Kalman filter, introduced here, plays a crucial role in control theory due to its ability to provide optimal and efficient state estimation for dynamic systems in the presence of noise.

I have also introduced the control problem and presented the Linear Quadratic Gaussian (LQG) problem as a significant application of the Kalman filter in control theory. The LQG problem exemplifies how the Kalman filter can be utilized to achieve optimal control in systems where both state estimation and control are necessary.

Furthermore, the relationship between martingales and the Kalman filter is explored, particularly in the context of steady-state scenarios, see section 2.6 in Akoi's *Optimization of Stochastic Systems: Topics in Discrete-Time Systems* (1989) [1]. Although the thesis begins with the simplest setting of the Kalman filter, where noise processes are independent and identically distributed (i.i.d.) Gaussian, it is shown that the filter' s and control's framework can be extended in connection with martingale difference sequences in Caines's *Linear Stochastic Systems* (1988) [8]. It provides a deeper understanding and broader applicability of the Kalman filter.

Here I state the extended setting with the martingale difference in Caines's *Linear Stochastic Systems* (1988, p.683) [8].

The system is described by

$$x_{k+1} = Fx_k + Gu_k + w_k, \quad k \in \mathbb{Z}_+,$$

$$y_k = Hx_k + Du_k + v_k, \quad k \in \mathbb{Z}_+,$$

with y, x, u, w, v defined on the probability space $(\Omega, \mathcal{B}, P)$, $x_k, w_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, $y_k, v_k \in \mathbb{R}^p$ for all $k \in \mathbb{Z}_+$, and all matrices appropriately conformable with these vectors. Let

$$\mathcal{F}_j = \sigma\left(x_0, \begin{bmatrix} w \\ v \end{bmatrix}_0, \ldots, \begin{bmatrix} w \\ v \end{bmatrix}_j\right), \quad j \in \mathbb{Z}_+, \quad \mathcal{F}_{-1} := \{\phi, \Omega\}$$

and assume $\begin{bmatrix} w \\ v \end{bmatrix}$ has the martingale difference property

$$\mathbb{E}\left[\begin{bmatrix} w \\ v \end{bmatrix}_k \middle| \mathcal{F}_{k-1}\right] = 0 \quad \text{a.s. } k \in \mathbb{Z}_1,$$

$$\mathbb{E}\left[\begin{bmatrix} w \\ v \end{bmatrix}_k \begin{bmatrix} w^T & v^T \end{bmatrix}_k \middle| \mathcal{F}_{k-1}\right] = \begin{bmatrix} O & O \\ O & R \end{bmatrix} \quad \text{a.s. } k \in \mathbb{Z}_1, \quad \mathbb{E}\left[\begin{bmatrix} w_0 \\ v_0 \end{bmatrix} x_0^T\right] = 0.$$

Here, purely for the sake of convenience, we have taken the cross-covariance between w_k and v_k to be zero at any instant k.

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Further let x_0 and the process $\begin{bmatrix} w \\ v \end{bmatrix}$ have a joint Gaussian distribution.

The control information pattern is given by

$$u_k \in \mathcal{U}_k := \left\{ u : \mathbb{R}^{m+p} \to \mathbb{R}^m, u_k \in \mathcal{G}_{k-1}^u \right\}$$

for $k \in \mathbb{Z}_1$, where

$$\mathcal{G}_k^u := \sigma(G^k) = \sigma\{y_0, \ldots, y_k, u_1, \ldots, u_k\} = \sigma\{y_0, \ldots, y_k\}, \quad k \in \mathbb{Z}_1,$$

$$\mathcal{G}_0^u := \{\phi, \Omega\},$$

so, in particular, $u_0 \in \mathbb{R}^m$ is an a.s. constant random variable.

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Appendix A

Appendix: Gaussian Space

A.1 Gaussian Space

The study of Gaussian spaces is fundamental in understanding the structure of various stochastic processes and their properties. This section introduces key concepts and results in Gaussian spaces, primarily drawing from Janson's comprehensive treatment in *Gaussian Hilbert Spaces* (1997, Chapter 1) [12], and LeGall's work in *Brownian Motion, Martingale, and Stochastic Calculus* (2016, Chapter 1) [13]. These sources provide the theoretical foundation upon which the above theorems and discussions are based.

A.1.1 Joint Normal

Definition A.1.1 (Joint normal)**.** *A finite number of random variables $X_1, \ldots, X_n$ are said to have a joint normal distribution, if $\sum_{i=1}^{n} a_i X_i$ is normal for any $a_1, \ldots, a_n \in \mathbb{R}$.*
For infinite number of random variables $(X_i)_{i \geq I}$ are of joint normal distribution if every finite subset of it has a joint normal distribution i.e.

$$\forall p \in \mathbb{N},\ i_1, \ldots, i_p \geq 1, \quad X_{i_1}, \ldots, X_{i_p} \quad \textit{is joint normal.}$$

Remark 8. *If $X_1, \ldots, X_n$ are independent normal variables, then $X_1, \ldots, X_n$ have joint normal distribution.*

Definition A.1.2. *A (centered) Gaussian space is a closed linear subspace of $\mathcal{L}^2(\Omega, \mathcal{F}, P)$ consisting of only centered Gaussian variables.*

More formally, consider $G \subseteq \mathcal{L}^2(\Omega, \mathcal{F}, P)$ to be a Gaussian space. If $(X_n)_{n\geq 1} \in G$ and X_n converges to X in $\mathcal{L}^2$. Then $X \in G$.

Theorem A.1.1. *Any set of random variables in a Gaussian space is of a joint normal distribution.*

Proof. If the set containing infinite number of random variables, by the definition just to verify that every finite subset of it is joint normal. So it's sufficient to consider a finite number of random variables.
Denote the Gaussian space by G. Let $X_1, \ldots, X_n \in G$ and $a_1, \ldots, a_n \in \mathbb{R}$. We have $\sum_{i=1}^{n} a_i X_i \in G$. (Since G is a linear space.) Thus $\sum_{i=1}^{n} a_i X_i$ is centered normal. So $X_1, \ldots, X_n$ is joint normal. □

A.1.2 Gaussian Spaces from Joint Normal

The Gaussian spaces generated by specific collections of random variables provide a practical approach to constructing and analyzing Gaussian spaces. We will now explore how finite and infinite sets of jointly normal random variables generate Gaussian spaces.

Theorem A.1.2. *Let $X_1, \ldots, X_n$ have a centered joint normal distribution. Then the linear span of $X_1, \ldots, X_n$*

$$span\{X_1, \ldots, X_n\} = \{\sum_{i=1}^{n} a_i X_i : a_i \in \mathbb{R}, 1 \leq i \leq n\}$$

is a finite-dimensional Gaussian space.

Theorem A.1.3. *Let $(X_n)_{n\geq 1}$ have a centered joint normal distribution. Then the closed linear span of $(X_n)_{n\geq 1}$,*

$$\overline{span}\{X_n : n \geq 1\} = \{\sum_{i=1}^{k} a_i X_i : a_i \in \mathbb{R}, 1 \leq i \leq k, \forall k \in \mathbb{N}\}$$

is a Gaussian space.

The above theorems provide a foundation for constructing Gaussian spaces from jointly normal variables, both finite and infinite. We will now proceed to discuss the relationship between independence and orthogonality in the context of Gaussian spaces.

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Remark 9 (Independence as Orthogonality)**.** *The connection between independence and orthogonality is particularly elegant in the setting of Gaussian spaces. Specifically, for centered Gaussian random variables X and Y in $\mathcal{L}^2$, independence is equivalent to orthogonality, meaning that $E[XY] = 0$ when X and Y are independent.*

Theorem A.1.4 (Conditional Expectation as Projection)**.** *Let G be a centered Gaussian space, and let K be a closed linear subspace of G. For any random variable $X \in G$, the conditional expectation $E[X \mid \sigma(K)]$ is equivalent to the orthogonal projection of X onto the subspace K within the Hilbert space $\mathcal{L}^2$. Formally:*

$$\mathbb{E}[X \mid \sigma(K)] = P_K(X),$$

where $P_K(X)$ represents the projection of X onto K. This projection is the best linear approximation of X from K, minimizing the expected squared difference $E[(X - P_K(X))^2]$.

Moreover, under this projection, the conditional distribution of X given $\sigma(K)$ remains Gaussian, with mean $P_K(X)$ and variance $\sigma^2 = E[(X - P_K(X))^2]$.

In Gaussian spaces, orthogonal projection onto a subspace K not only provides the optimal linear estimate of X based on the information contained in $\sigma(K)$, but also preserves the Gaussian nature of the distribution, which is crucial in many applications such as the Kalman filter.

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